



FNQROC DEVELOPMENT MANUAL

CP1- Appendices

CONSTRUCTION PROCEDURES - Appendices

Version No. 05/23

This work is copyright. Apart from any use as permitted under the Copyright Act 1968, no part may be reproduced by any process without prior written permission from the Far North Queensland Regional Organisation of Councils. Requests and inquiries concerning reproduction and rights should be addressed to the FNQROC Coordinator, PO Box 359, CAIRNS, Qld 4870.

TABLE OF CONTENTS

APPENDIX	TITLE	PAGE
APPENDIX A - Inspection And Test Requirements.....		1
APPENDIX B – Sewer Mains Inspection and Test Plan Template		13
APPENDIX C – Water Main Inspection and Test Plan Template		22
APPENDIX D – Pump Station Inspection and Test Plan Template		33
APPENDIX E – Security Lodgement Form.....		44
APPENDIX F - Inspection Certificate for Witness/Hold Points		45
APPENDIX G – Works Acceptance Inspection Checklist.....		46
APPENDIX H - Final Acceptance Inspection Checklist		56
APPENDIX I - Sewerage and Water Pump Station Commissioning Checklists ^{CRC}		61
PRE-COMMISSIONING CHECKLIST FOR NEW ASSETS WATER SITES		61
APPENDIX I - Sewerage and Water Pump Station Commissioning Checklists ^{CRC}		68
APPENDIX J - Statement of Compliance “As Constructed”		80
APPENDIX K - Example of Surveyor Certification of “As Constructed” Works.....		81
APPENDIX L As Constructed Data Sewer Property Connection Branches.....		82
APPENDIX M - As Constructed Data Submission Form.....		84
APPENDIX N - (Pending) Permanent Survey Mark Coordination Request Form		85
APPENDIX O - Deleted		86
APPENDIX P - “As Constructed” Digital Data and Drawing Standards		87

APPENDIX A - Inspection And Test Requirements

Elements of Work	Consulting Engineer's Responsibility	Council's Responsibility
Clearing and Grubbing Location Allotment Filling Quality of Material Compaction Alignment	HOLD POINT Upon completion of survey, inspect defined limits of clearing. Examine and assess all test results. Examine and assess all test results.	WITNESS POINT Joint inspection of defined limits and tree removal if considered warranted. Visit site for random audit inspections if considered warranted. Visit site for random audit inspections if considered warranted. Visit site for random audit inspections if considered warranted.
Subgrade Compaction CBR Tests (if ordered) Horizontal and Vertical Alignments Profile Embankments	Routinely visit site. HOLD POINT Attend during proof rolling. Examine and assess all test results. HOLD POINT Where CBR tests dictate an alternative pavement design, the council approval of a revised pavement design shall form a hold point. Routinely visit site. Examine and assess all test results and cross section geometry. Routinely visit site. HOLD POINT Attend during completion of trimming and removal of soft spots. Routinely visit site. HOLD POINT Attend during placement of fill.	Visit site for random audit inspections HOLD POINT Joint inspections during proof rolling. Assess all test results. HOLD POINT Where CBR tests dictate an alternative pavement design, the approval of a revised pavement design shall form a hold point. Visit site for random audit inspections Assess all test results and cross section geometry. Visit site for random audit inspections HOLD POINT Joint inspections during trimming and removal of soft spots. WITNESS POINT Visit site for random audit inspections if considered warranted
Subgrade Replacement Material Quality Compaction (a) For on site material (b) For graded material Profile & Depth	HOLD POINT Make sufficient routine visits to assess quality of materials. HOLD POINT Examine and assess all test results. HOLD POINT Make sufficient routine visits to assess that operations will achieve a sound compacted layer HOLD POINT Examine and assess all test results. HOLD POINT Examine and assess all test results.	WITNESS POINT Visit site for random audit inspections if considered warranted. WITNESS POINT Visit site for random audit inspections if considered warranted. WITNESS POINT Visit site for random audit inspections if considered warranted.
Building/Structures	Ensure Council Approval of all building/structures	Inspect and ensure compliance

CONSTRUCTION PROCEDURES – Appendices

Elements of Work	Consulting Engineer's Responsibility	Council's Responsibility
Sub-base Layer Material Quality Compaction Profile & Depth	Routinely visit site. Examine and assess all test results. Routinely visit site. HOLD POINT Attend during proof rolling. Examine and assess all test results. Routinely visit site. HOLD POINT Attend at completion of final preparation. Examine and assess all test results and cross section geometry.	WITNESS POINT Visit site for random inspections if considered warranted. WITNESS POINT Visit site for random inspections if considered warranted. HOLD POINT Joint inspection on completion of final preparation.
Base Layer Material Quality Compaction Horizontal and Vertical Alignments (a) With no Kerb & Channelling (b) With Kerb & Channelling Profile	Routinely visit site. Routinely visit site. HOLD POINT Attend during proof rolling Examine and assess all test results and cross section geometry. Routinely visit site. Examine and assess all test results and cross section geometry. Routinely visit site. HOLD POINT Attend at completion of final preparation.	Visit site for random audit inspections Visit site for random audit inspections HOLD POINT Joint inspections during proof rolling. Visit site for random audit inspections Visit site for random audit inspections HOLD POINT Joint inspection on completion of final preparation.
Surfacing Material Quality Compaction and Thickness Horizontal & Vertical Alignments Profile	 HOLD POINT Undertake a Pre-seal Inspection.	WITNESS POINT Visit site for random inspections if considered warranted WITNESS POINT Visit site for random inspections if considered warranted WITNESS POINT Visit site for random inspections if considered warranted HOLD POINT Undertake a Pre-seal Inspection
Subsoil Drains Pipe Filter Material Cleaning Joints and Markers Geofabric	Routine inspections of Contractors Performance and progress of works. Routine inspections of Contractors Performance and progress of works. Routine inspections of Contractors Performance and progress of works. Routine inspections of Contractors Performance and progress of works.	Visit site for random inspections if considered warranted Visit site for random inspections if considered warranted Visit site for random inspections if considered warranted Visit site for random inspections if considered warranted
Kerb and Channel Material Quality Horizontal & Vertical Alignments	HOLD POINT Inspect foundations prior to kerb placement Inspect completed kerb. Water test where appropriate.	HOLD POINT Visit site for inspection HOLD POINT Visit site for inspection

CONSTRUCTION PROCEDURES – Appendices

Elements of Work	Consulting Engineer's Responsibility	Council's Responsibility
Road Crossing Conduits Location Markers	Routine inspections of Contractors Performance. Routine inspections of Contractors Performance.	Visit site for random inspection is considered warranted. Visit site for random inspection is considered warranted.
Stormwater Drainage Location of Structures SL & IL at Structures Material Quality (Bedding, Concrete, Pipes) Manholes Drain lines Backfilling 'Cast Insitu' Reinforced Concrete Work	HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Visual inspection prior to placement of structure/s after bedding sand. HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Sufficient visits to assess compliance and to view progress and works. HOLD POINT Inspect reinforcement and formwork prior to concrete pour	 WITNESS POINT Visit site for inspection prior to laying of pipe and bedding. HOLD POINT Visit site for inspection prior to backfilling. WITNESS POINT Inspect reinforcement and formwork prior to concrete pour
Landscaping Grass establishment Tree Planting Irrigation (a) Pipelines (b) Pressure Testing Pipelines (c) Performance Testing	Routine inspections of Contractors Performance. Routine inspections of Contractors Performance. Confirm all affected areas are topsoiled, grassed and maintained. WITNESS POINT Witness and approve pressure and performance test.	Visit site for check at defects liability inspection.
Soil and Water Quality	HOLD POINT Examine and approve contractors ESCP for compliance with ESCS. ESC Measures for works area are in place prior to works commencing on this section or stage. Randomly audit and inspect ESC measures for compliance with contractors ESCP.	WITNESS POINT Visit site for inspection if considered warranted.

CONSTRUCTION PROCEDURES – Appendices

Elements of Work	Consulting Engineer's Responsibility	Council's Responsibility
Sewerage Reticulation Location MH's & HC's IL at MH & HC's Backfilling SP Boundary Set Out Material Quality (Bedding, Concrete, Pipes) Manholes, Maintenance Shafts & Benching Hydrostatic Testing of Manholes	Routine inspections Routine inspections and review of field information Routine inspections of Contractors Performance . HOLD POINT Visual inspection after excavation prior to bedding. Routine inspections of Contractors performance. Review of field measurements. Assess all test results. Routine inspections HOLD POINT Witness hydrostatic testing of manholes.	 WITNESS POINT Visual inspection after excavation prior to bedding. Visual inspection of lines prior to backfill. WITNESS POINT Witness hydrostatic testing of manholes.
Pipelines Thrust/anchor blocks Trunk Infrastructure	HOLD POINT Witness pressure test of lines . . HOLD POINT Visual site inspection of anchor/thrust blocks prior to concrete pour. HOLD POINT Pre-connection visual inspection of trunk lines.	HOLD POINT Visual inspection after excavation prior to bedding. HOLD POINT Witness pressure test of lines. WITNESS POINT Visual site inspection of anchor/thrust blocks prior to backfill. HOLD POINT Pre-connection visual inspection of trunk lines.
Water Reticulation Location SP Boundary Set Out Valves, Hydrants, Scours, Bends Depth Material Quality (Bedding, Concrete, Pipes) Pipelines Thrust/anchor blocks Backfilling	Routine inspections of Contractors Performance. Review of field measurements Routine inspections of Contractors Performance. Review of field measurements Routine inspections of Contractors Performance. Review of field measurements Routine inspections of Contractors Performance. Review of field measurements Assess all test results. HOLD POINT Visual inspection after excavation prior to bedding. HOLD POINT Witness pressure test of lines. HOLD POINT Witness chlorine swabbing of lines – pre-amalgamation DSC area. HOLD POINT Disinfection (if directed) / flush of pipeline HOLD POINT Visual site inspection of anchor/thrust blocks prior to concrete pour. Routine inspections of Contractors Performance . HOLD POINT Visual site inspection prior to backfill	 WITNESS POINT Visual inspection after excavation prior to bedding. HOLD POINT Witness pressure test of lines. WITNESS POINT Disinfection / flush of pipeline WITNESS POINT Visual site inspection of anchor/thrust blocks prior to backfill. Visual inspection of lines prior to backfill. WITNESS POINT Visual site inspection prior to backfill
Elements of Work	Consulting Engineer's Responsibility	Council's Responsibility

CONSTRUCTION PROCEDURES – Appendices

Pump Stations and Valve Chambers Excavation Foundation Inspection Base slab reinforcement, formwork and water stop Reinforcement and formwork Materials testing Hydrostatic Testing Electrical and SCADA equipment Lifting chain Pump testing and Station Commissioning	Routine inspections of Contractor's performance WITNESS POINT. Confirm water table level and founding condition WITNESS POINT Inspect foundation prior to placing formwork / reinforcement HOLD POINT Inspect reinforcement, water stop and formwork prior to concrete pour. HOLD POINT Inspect reinforcement and formwork prior to concrete pour. Assess all test results. HOLD POINT Witness hydrostatic test. WITNESS POINT. Review switchboard test certification. Inspect installation WITNESS POINT Review certification of lifting chain. HOLD POINT Witness pressure and draw down testing of pumps. HOLD POINT Inspection against commissioning checklist.	WITNESS POINT Inspect foundation prior to placing formwork / reinforcement Visual inspection of reinforcement, water stop and formwork prior to concrete base pour. Visual inspection of reinforcement and formwork prior to concrete pour WITNESS POINT Visual inspection of concrete prior to stripping of formwork. WITNESS POINT Review materials testing. HOLD POINT Witness hydrostatic test. HOLD POINT ¹Witness factory acceptance testing of switchboards prior to delivery to site (¹ At receiving council's discretion) WITNESS POINT Review certification of lifting chain. HOLD POINT Witness pressure and draw down testing of pumps. HOLD POINT Inspection against commissioning checklist.
Prior to Acceptance of works for "Defects Liability Period"	Forward "As Constructed" submission to Council with Surveyor and Consulting Engineers certification attached. Finalise all other Documentation in accordance with Construction Procedures. Complete "Defects Liability" Inspection Checklist prior to joint inspection with Council	Council to accept and conduct Audit checks of As Constructed Drawings and advise any requirements. Council Inspector to accompany Consulting Engineer and Contractor and to advise any requirements. When completed advise in writing of acceptance of works for commencement of "Defects Liability Period"
During "Defects Liability Period"	Consulting Engineer to confirm all minor omissions and defects have received suitable attention and to examine and approve site prior or asking for "Final Acceptance of works" Inspection	Council to advise Consulting Engineer of any defects
Prior to Final Acceptance of works.	Consulting Engineer to accompany Council Inspector and to note any requirements	Council Inspector to accompany Consulting Engineer and Contractor and to advise any requirements. When completed advise in writing of final acceptance of works.

TEST REQUIREMENTS

Construction Activity	Verification Requirement		Minimum Test Frequency	Specification Requirement	Minimum No. of Tests	
	Description	Test Required				
SEWER MAIN CONSTRUCTION						
Embedment	Compaction	WSA02-2014				
Trench Fill	Compaction	WSA02-2014				
Gravity Pipes	Air Pressure and Vacuum	Table S6.2, Operational Works Specification S6 “Sewerage Reticulation”, FNQROC Development Manual				
	Deflection	WSA02-2014				
	CCTV Inspection	WSA02-2014				
Manholes	Vacuum or Hydrostatic	Clause S6.26, Operational Works Specification S6 “Sewerage Reticulation”, FNQROC Development Manual				
SEWER PUMP STATION CONSTRUCTION						
Embedment	Compaction	WSA04-2005 36.3				
Backfilling	Compaction	WSA04-2005 36.3				
Switchboards	Electrical Testing	WSA04-2005 36.9				
WATER MAIN CONSTRUCTION						
Embedment	Compaction	WSA03-2002 19.3				
Trench Fill	Compaction	WSA03-2002 19.3				
Pipes	Pressure	Clause S5.28, Operational Works Specification S5 “Water Reticulation”, FNQROC Development Manual				
Disinfection (if directed)	Bacteriological	WSA03-2002 19.5				
STORMWATER DRAINAGE CONSTRUCTION						
Excavation	Compaction	RDD	Q141B/Q140A/Q143 or	1/50m²	95% SRDD	1
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD	n/a	
			AS 1289.5.1.1 or			
			AS 1289.5.7.1			

Construction Activity	Verification Requirement			Minimum Test Frequency	Specification Requirement	Minimum No. of Tests
	Description	Test Required				
STORMWATER DRAINAGE CONSTRUCTION (cont'd)						
Bedding/Haunch (RCP, RCBC or similar)	Material Quality	Grading	Q103A or	1/material type	Table 19.2.6, MTRS04	n/a
			AS 1289.3.6.1			
		Linear Shrinkage	Q106 or			
			AS 1289.3.4.1			
	Compaction (Cohesive)	RDD	Q141B/Q140A/Q143 or	Under trafficable area 1/side/50m	Table S4.1, Operational Works Specification S4 "Stormwater Drainage", FNQROC Development Manual	2
			AS 1289.5.3.1	Elsewhere 1/side/100m		
		MDD	Q141A or	1/material type		n/a
			Q142E			
	Compaction (Cohesionless)	Density Index	Q132B or	Under trafficable area 1/side/50m		2
			AS 1289.5.6.1	Elsewhere 1/side/100m		
		Min/Max Dry Density	AS 1289.5.1.1 or	1/material type	n/a	
			AS 1289.5.5.1			
Backfill (RCP, RCBC or similar)	Material Quality	Grading	Q103A or	1/material type	Table 19.2.3, MRTS04	n/a
			AS 1289.3.6.1			
		Linear Shrinkage	Q106 or			
			AS 1289.3.4.1			
	Compaction (Design Trench Width ≤ 4m)	RDD	Q141B/Q140A/ Q143 or	Under trafficable area 1/300mm lift/50m	Table S4.1, Operational Works Specification S4 "Stormwater Drainage", FNQROC Development Manual	1 (between structures)
			AS 1289.5.3.1 or	Elsewhere 1/900mm lift/100m		
			AS 1289.5.4.1			
		MDD	Q110A or	1/material type		n/a
			AS 1289.5.1.1 or			
			AS 1289.5.7.1	1/RDD		

Construction Activity	Verification Requirement			Minimum Test Frequency	Specification Requirement	Minimum No. of Tests	
	Description	Test Required					
STORMWATER DRAINAGE CONSTRUCTION (cont'd)							
Backfill (RCP, RCBC or similar) (cont'd)	Compaction (Design Trench Width > 4m)	RDD	Q141B/Q140A/Q143 or	Under trafficable area 1/300mm lift/200m ² Elsewhere 1/900mm lift/400m ²		1 (between structures)	
			AS 1289.5.3.1 or				
			AS 1289.5.4.1				
		MDD	Q141A or	1/material type		n/a	
			AS 1289.5.1.1 or				
			AS 1289.5.7.1				1/RDD
Backfill (In-Place Structures other than RCP, RCBC or similar)	Material Quality	Grading	Q103A or	1/material type	100% < 50mm	n/a	
			AS 1289.3.6.1		2 ≤ IP ≤ 12		
		Plasticity index	Q105 or			Table S4.1, Operational Works Specification S4 "Stormwater Drainage", FNQROC Development Manual	2
			AS 1289.3.3.1				
	Compaction	RDD	Q141B/Q140A/ Q143 or	2/500mm lift			
			AS 1289.5.3.1 or				
			AS 1289.5.4.1				
		MDD	Q141A or	1/material type			
			AS 1289.5.1.1 or				
			AS 1289.5.7.1			1/RDD	
Backfill (Stabilised Sand)	Material Quality	Stabilised sand shall comprise sand meeting the requirements of Table 19 MRS11.04 in an intimate mixture of 12 parts sand and 1 part of either Type GP or GB cement			Table 19.2.5, MRTS04	n/a	
Bedding/Haunch/Backfill/Overlay (Buried Metal Corrugated Structures)	Material Quality	All materials shall be in accordance with the manufacturer's recommendations. Evidence of these recommendations and subsequent compliance shall be incorporated with the Contractor's quality records.			As per manufacturer's recommendations.		
	Installation	Installation shall be in accordance with the manufacturer's recommendations. Evidence of these recommendations and subsequent compliance shall be incorporated with the Contractor's quality records			As per manufacturer's recommendations.		

Construction Activity	Verification Requirement		Minimum Test Frequency	Specification Requirement	Minimum No. of Tests
	Description	Test Required			
STORMWATER DRAINAGE CONSTRUCTION (cont'd)					

Stormwater Drainage System	CCTV Inspection	Clause S4.26, Operational Works Specification S4 “Stormwater Drainage”, FNQROC Development Manual			
----------------------------	-----------------	---	--	--	--

ROAD CONSTRUCTION						
Ground Surface Treatment	Compaction	RDD	Q141B/Q140A/Q143 or	1/2500m²	>0.3m below pavement subgrade - 95% SRDD	3
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD	<0.3m below pavement subgrade - 97% SRDD	n/a
			AS 1289.5.1.1 or			
			AS 1289.5.7.1			
Embankment (Road)	Compaction	RDD	Q141B/Q140A/Q143 or	1/200mm lift/2500m² or 1/500m³	>0.3m below pavement subgrade - 95% SRDD	3
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD	<0.3m below pavement subgrade - 97% SRDD	n/a
			AS 1289.5.1.1 or			
			AS 1289.5.7.1			
Embankment (Concentrated Operations – Gullies etc)	Compaction	RDD	Q141B/Q140A/Q143 or	1/200mm lift/500m² or 1/100m³	>0.3m below pavement subgrade - 95% SRDD	3
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD	<0.3m below pavement subgrade - 97% SRDD	n/a
			AS 1289.5.1.1 or			
			AS 1289.5.7.1			

Construction Activity	Verification Requirement			Minimum Test Frequency	Specification Requirement	Minimum No. of Tests
	Description	Test Required				
ROAD CONSTRUCTION (cont'd)						
Subgrade (General)	Material Quality	CBR	Q113C (soaked)	Representative each material and 1 test per 500m carriageway or part thereof	97% MDD 100% OMC	n/a
	Compaction	RDD	Q141B/Q140A/Q143 or	1/1000m²	97% SRDD	3
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD		n/a
			AS 1289.5.1.1 or			
AS 1289.5.7.1						
Subgrade (Turnouts and Entrances)	Compaction	RDD	Q141B/Q140A/Q143 or	1/100m²	97% SRDD	1
			AS 1289.5.3.1 or			
			AS 1289.5.4.1			
		MDD	Q141A or	1/RDD		n/a
			AS 1289.5.1.1 or			
			AS 1289.5.7.1			
Pavement Layers (General)	Material Quality	All materials shall be sourced from a Quality Assured material supplier and the results of the manufacturer's testing to assure quality of the product shall be incorporated with the Contractor's quality records			MRTS05 Section 7.2 "Type 2 Unbound Material"	
	Compaction	RDD	Q141B/Q140A/Q143 or	1/500m²	100% SRDD	4
			Q141A or	2/500m²		8
			AS 1289.5.3.1	1/500m²(2/500m² if using AS1289.5.8.1)		4 (8)
		MDD	Q141A or	1/material type/5000m²		n/a
			AS 1289.5.1.1 or			
			AS 1289.5.4.2			

Construction Activity	Verification Requirement		Minimum Test Frequency	Specification Requirement	Minimum No. of Tests	
	Description	Test Required				
ROAD CONSTRUCTION (cont'd)						
Pavement Layers (Turnouts and Entrances)	Compaction	RDD	Q141B/Q140A/Q143 or	1/100m²	100% SRDD	1
			Q141A or	2/100m²		
			AS 1289.5.3.1	1/100m²(2/100m² if using AS1289.5.8.1)		
		MDD	Q141A or	1/material type	n/a	
			AS 1289.5.1.1 or			
			AS 1289.5.4.2			as required
Structural Concrete	Compressive Strength		AS 1012.1	1 sample of 2 cylinders for each 15m³ or part thereof placed in an essentially continuous manner	Table S7.1, “Concrete Classes”, FNQROC Development Manual	1 sample per casting day
			AS 1012.3.1			
			AS 1012.8.1			
			AS 1012.9			
			AS 1012.12.1			

CONSTRUCTION PROCEDURES – Appendices

Table 19.2.3 - Select Backfill Properties

MRTS04

AS SIEVE SIZE (mm)	Percent (by mass) Passing Sieve	
	Gravel †	Loam
37.5	100	100
9.5	60 – 85	100
2.36	25 - 70	70 – 100
0.425	10 – 40	10 – 40
0.075	3 - 30	3 - 30
Other Properties Linear Shrinkage	8 maximum	6 maximum

† Material of size greater than 2.36mm shall be stone

Table 19.2.5 - Sand Properties

MRTS04

Property	Natural Sand	Blended and Manufactured Sand
Percent passing 6.7mm AS sieve	100	100
Percent passing 0.075mm AS sieve (maximum)	5	20
Plasticity Index (maximum)	5	10

Table 19.2.6 – Grading Limits for Bedding Material

MRTS04

AS SIEVE SIZE (mm)	% Passing By Mass
19	100
2.36	30 – 100
0.425	15 – 70
0.075	Mar-30
Other Properties Linear Shrinkage	6 maximum

APPENDIX B – Sewer Mains Inspection and Test Plan Template

INSPECTION AND TEST PLAN – SEWER MAIN LAYING

Developer:		Consultant Engineer:		Consultant Engineer Representative:					
Project:		Contractor:		Contractor Site Representative:					
Description:		Sub-contractor:		Witness, Hold & Surveillance points added to ITP					
		Field Tester:							
Location:		ITP Prepared by:		Reviewed by:		Council Representative:			
		Date / /		Date / /					
No	Construction/Inspecti on Activity	Inspection Procedure & Acceptance Criteria			Cont-ractor	Consulti ng Engineer	Council *	Record	Comment
1	Pre start/Site establish	Pre-Start Meeting Checklist. Site establishment visual check. Checklist completed. (S1)			I	H	H	Checklist S1	
2	Approved materials on Site and delivered	Visual check approved materials. Quantity and condition. Checklist completed (S2)			I	I	S	Checklist S2	
3	Excavation and pipe laying	Visual and dimensional check to FNQROC Standards. Checklist completed. (S3)			I	H	W	Checklist S3	
4	Maintenance Shafts	Visual and dimensional check to FNQROC Standards. Checklist completed (S4)			I	H	S	Checklist S4	
5	Survey of main in easement	Survey of pipe location before backfill			I	I	S		
6	Anchor/Thrust Blocks	Visual and dimensional check to FNQROC Standards. Checklist completed.			I	H	H		
7	Embedment and Trench Fill	Visual check and compaction testing to FNQROC Standards			I	H	W	Compaction test results	
8	Surface fittings	Visual and dimension check to FNQROC Standards (S5)			I	I	S	Checklist S5	
9	Testing	Pressure/vacuum, deflection and compaction tests to FNQROC Standards			I	H	W	Test Results	
10	Pre-Connection Inspection	Visual inspection to all parties standards (S6) Flow Management procedures as per Job Specific Schedule Letter			H	H	H	Checklist S6	
12	Restoration	Visual inspection against photographs. Clearance letter from Council/property owner			I	W	S	Clearance letter	
Symbol	Legend	No	Amendment		Date	Reviewed	Validation		
I	Inspection						I certify that the works have been constructed in accordance with FNQROC Standards and the Inspection and Test Plan Consulting Engineer Date / /		
H	Mandatory Hold Point								
W	Witness Point								
S	Surveillance								

* Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements

SEWER CHECKLIST S1
PRE-START AND SITE ESTABLISHMENT

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
1.1	Plan current and on site												
1.2	Pre construct report inc. photographs												
1.3	Road opening requirements												
	Fees paid							Council Requirements					
	Traffic mgt plan implemented												
1.4	Environmental Management Plan on site and implemented												
1.5	Safe Work Plan on site and implemented												
1.6	Receiving sewer located												
1.7	Property entry agreement												
1.8	Main laying Specification on site												
1.9	Footways to finished levels							If laying in road reserve					
1.10	Survey pegs in place							Surveyor					
1.11	Job set out/level sheets												
1.12	All services located							'Dial Before You Dig', Services Search, and Relevant Authorities					
1.13	All services marked												
1.14	Main layer holding relevant accreditation on site												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

SEWER CHECKLIST S2 APPROVED MATERIALS ON SITE AND DELIVERED											
										* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL	
PROJECT:								CONSULTING ENGINEER:			
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:			
Date from: to:		DAY				DATE DAY 1:		DAY 2:	DAY 3:	DAY 4:	DAY 5: DAY 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures
2.1	Delivery Inspection										
2.2	Pipe type and size to current plan							Current Plan			
2.3	Bedding material										
2.4	Trench fill										
2.5	Fittings										
2.6	Surface Fittings										
2.7	Pre Cast chambers										
2.8	Mortar / Plaster										
2.9	Pipe laying accessories										
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:			
COMMENT:											

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

SEWER CHECKLIST S3 – PAGE 1 OF 2
EXCAVATION AND PIPELAYING

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		DAY						DATE DAY 1:	DAY 2:	DAY 3:	DAY 4:	DAY 5:	DAY 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
3.1	Environmental Management Plan on site and implemented												
3.2	WH&S Plan on site and implemented												
3.3	Services exposed												
3.4	Clearance from Services												
3.5	Trench width mm												
3.6	Trench depth to design level												
3.7	Trench shoring/Plating												
3.8	Trench drainage												
3.9	Pipe Embedment												
	Compaction												
	Bedding mm												
	Surrounds/sides mm												
3.10	Pipe laid to grade												
3.11	Jointing to Standards												
3.12	Concrete												
	Trench stops in place												
	Bulkheads in place												
3.13	Property Connection Sewers to Standard												
	Identification tape												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

SEWER CHECKLIST S3 – PAGE 2 OF 2
EXCAVATION AND PIPELAYING

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		DAY						DATE DAY 1:	DAY 2:	DAY 3:	DAY 4:	DAY 5:	DAY 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
3.14	Trench Fill												
	Material												
	Compaction												
	Compaction test							NATA Certified lab					
3.15	Terminal Maintenance Shaft to Standard												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

**SEWER CHECKLIST S4
MAINTENANCE SHAFTS**

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:										CONSULTING ENGINEER:									
PIPE TYPE:										CLASS:									
SIZE:										CONTRACTOR:									
Date from: to:										CHAMBER									
										DATE CH 1: CH 2: CH 3: CH 4: CH 5: CH 6:									
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD				COMMENT				Signatures			
4.1	Finished Surface Levels Supplied																		
4.2	Drainage requirements																		
4.3	Base																		
	Placement																		
	Channels																		
	First shaft section																		
4.4	Pre cast chamber																		
	Shaft assembled in correct order																		
	Step iron location and spacing																		
	Sealing																		
	Offset cone located correctly																		
	Minimum one make up ring																		
	Cover and frame																		
4.5	In-situ chamber																		
	Reinforcement							Plan Specification											
	Cover																		
	Concrete type to Specification																		
	Step iron location and spacing																		
	Dimension check																		
	Cover and frame																		
4.6	Plastering/rendering																		
4.7	Benching																		
VARIATIONS AND CHANGES:										SITE INSTRUCTIONS:									
COMMENT:																			

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

SEWER CHECKLIST S5
SURFACE FITTINGS

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
5.1	Surface boxes and surrounds to finished levels												
5.2	Surface box lids hinged in direction of traffic flow												
5.3	Shroud pipes assembled to Standards												
5.4	Fitting bolts protected to Standards												
5.5	Correct depth to Spindle tops												
5.6	Correct depth to Hydrant lugs												
5.7	Spindle retaining disc in place												
5.8	Indicator plates in place												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

SEWER CHECKLIST S6
PRE-CONNECTION INSPECTION

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:										CONSULTING ENGINEER:												
					PIPE TYPE:			SIZE:			CLASS:			CONTRACTOR:								
Date from: to:					SITE						DATE SITE 1:		SITE 2:		SITE 3:		SITE 4:		SITE 5:		SITE 6:	
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD				COMMENT				Signatures						
6.1	WAC compiled																					
6.2	Compaction and concrete tests																					
6.3	Vacuum/Air test results																					
6.4	Deflection Test Results																					
6.5	CCTV Inspection																					
6.6	Junction tracer tape in place																					
6.7	Surface boxes and surrounds level																					
6.8	Terminal Mtce Shaft/Rodding Points to Standard																					
6.9	Chambers sized to Standard																					
6.10	Cover and frame to Standard																					
6.11	Minimum one make up ring																					
6.12	Chamber step irons to Standard																					
6.13	First shaft section																					
6.14	Channels to Standard																					
6.15	Benching to Standard																					
6.16	Sealing to Standard																					
6.17	Site restored satisfactorily																					
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:														
COMMENT:																						

*** Council reserves the right to vary these requirements at any time ** Council's written approval MUST be obtained prior to varying these requirements**

APPENDIX C – Water Main Inspection and Test Plan Template

INSPECTION AND TEST PLAN – WATER/RECYCLED WATER MAIN LAYING

Developer:		Consultant Engineer:		Consultant Engineer Representative:				
Project:		Contractor:		Contractor Site Representative:				
Description:		Sub-contractor:		Witness, Hold & Surveillance points added to ITP				
		Field Tester:						
Location:		ITP Prepared by:		Reviewed by:		Council Representative		
		Date / /		Date / /				

No	Construction/Inspecti on Activity	Inspection Procedure & Acceptance Criteria	Cont- ractor	Consult. Engineer	Council *	Record	Comment
1	Pre-start/Site establish	Pre-Start Meeting Checklist. Site establishment visual check. Checklist completed. (W1)	I	H	H	Checklist W1	
2	Approved materials on Site/delivered	Visual check approved materials. Quantity and condition. Checklist completed (W2)	I	I	S	Checklist W2	
3	Excavation & Pipe Laying	Visual and dimensional check to FNQROC Standards. Checklist completed. (W3)	I	H	W	Checklist W3	
4	Chambers	Visual and dimensional check to FNQROC Standards. Checklist completed (W4)	I	I	S	Checklist W4	
5	Survey of main in easement	Survey of pipe location before backfill	I	I	S		
6	Anchor/Thrust Blocks	Visual and dimensional check to FNQROC Standards. Checklist completed.	I	H	W		
7	Embedment and Trench Fill	Visual check and compaction to FNQROC Standards	I	H	W	Compaction test results	
8	Surface fittings	Visual and dimension check to FNQROC Standards. Checklist completed (W5)	I	I	S	Checklist W5	
9	Disinfection	Disinfection to FNQROC Standards	I	H	W	Test Results	
10	Testing	Pressure test and Compaction test to FNQROC Standards	I	H	H	Test Results	
11	Pre-connection inspection	Visual inspection to FNQROC Standards. Checklist completed (W6, W7) Isolation procedure as per Job Specific Letter	H	H	H	Checklist W6, W7	
12	Dual water flow test property service/s (main to meter)	Flow test of drinking water system. Lock and tag meter ball valve. Flow test of recycled water system. Lock and tag meter ball valve.	I	W	H	Test results	
13	Restoration	Visual inspection against photographs. Clearance letter from Council/property owner	I	W	S	Clearance letter	

Symbol	Legend	No	Amendment	Date	Reviewed	Validation
I	Inspection					I certify that the works have been constructed in accordance with FNQROC Standards and the Inspection and Test Plan Consulting Engineer Date / /
H	Mandatory Hold Point					
W	Witness Point					
S	Surveillance					

* Council reserves the right to vary these requirements at any time

** Council's written approval MUST be obtained prior to varying these requirements

WATER CHECKLIST W1 PRE-START AND SITE ESTABLISHMENT													
										* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL			
PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD		COMMENT		Signatures	
1.1	Plan current and on site												
1.2	Pre construct report inc. photographs												
1.3	Property Entry Agreement												
1.4	Road opening requirements												
	Fees paid												
	Traffic mgt plan implemented												
1.5	Environmental Management Plan on site and implemented												
1.6	Wh&S Plan on site and implemented												
1.7	Main laying Specification on site												
1.8	Footways to finished levels												
1.9	Survey pegs in place							Surveyor					
1.10	Job set out												
1.11	All services located							'Dial Before You Dig', services search and Relevant Authorities					
1.12	All services marked												
1.13	Main layer holding relevant accreditation on site												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

WATER CHECKLIST W2 APPROVED MATERIALS ON SITE AND DELIVERED													
										* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL			
PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD		COMMENT		Signatures	
2.1	Delivery Inspection												
2.2	Pipe type and size to current plan												
2.3	Polyethylene sleeving & accessories												
2.4	Marking tape												
2.5	Bedding material												
2.6	Trench fill												
2.7	Fittings												
2.8	Surface Fittings												
2.9	Pre Cast chambers												
2.10	Pipe laying accessories												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

WATER CHECKLIST W3 – PAGE 1 OF 2
EXCAVATION AND PIPE LAYING

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		DAY						DATE DAY 1:	DAY 2:	DAY 3:	DAY 4:	DAY 5:	DAY 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
4.1	Environmental Management Plan on site and implemented												
4.2	Traffic Management Plan on site and implemented												
4.3	Services exposed												
4.4	Clearance from Services												
4.5	Trench width mm												
4.6	Trench depth mm												
4.7	Trench shoring												
4.8	Excavation prior to placement of backfill								COUNCIL HOLD POINT				
4.9	Sleeving												
	Pipe and/or Fitting clean												
	Sleeving overlapped & sealed												
	Sleeving Ends sealed												
4.10	Laying												
4.11	Pipe Embedment												
	Compaction												
	Bedding												
	Surround												
	Overlay												
	Testing												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

WATER CHECKLIST W3 – PAGE 2 OF 2
EXCAVATION AND PIPE LAYING

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:				
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:				
		DAY				DATE DAY 1:		DAY 2:	DAY 3:	DAY 4:	DAY 5:	DAY 6
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures	
4.12	Pipe joints											
	Witness mark											
	Deflection limits											
	Restrained joints											
4.13	Valves, Hydrants & Surface fittings installed											
	Shroud assembly											
	Valve anchorage											
4.14	Marking tape											
	Correct location											
	Connected to fittings											
4.15	Concrete											
	Trench stops in place											
	Bulkheads in place											
	Thrust blocks in place								COUNCIL HOLD POINT			
	Embedment & Encasement in place											
4.16	Trench fill											
	Material											
	Compaction											
	Compaction Testing							NATA Certified Lab				
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:				
COMMENT:												

**WATER CHECKLIST W4
CHAMBERS**

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		CHAMBER						DATE CH 1:	CH 2:	CH 3:	CH 4:	CH 5:	CH 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
5.1	In-Situ Chamber												
	Formwork – correct sizing												
	Formwork – correct levels												
	Reinforcement												
	Conduits												
5.2	Pre Cast Chamber												
	Size to Standards												
	Base slab to Standards												
	Base levelled												
5.3	Reduced Size Chamber												
5.4	Scour Chamber												
5.5	Ladders / handrails / step irons												
5.6	Puddle Flanges												
5.7	Sealing							Manufacturer Specification					
5.8	Drainage												
5.9	Metal access cover												
5.10	Operational access												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

**WATER CHECKLIST W5
SURFACE FITTINGS**

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
6.1	Surface boxes and surrounds to finished levels												
6.2	Surface box lids hinged in direction of traffic flow												
6.3	Shroud pipes assembled to Standards												
6.4	Fitting bolts protected to Standards												
6.5	Correct depth to Spindle tops												
6.6	Correct depth to Hydrant lugs												
6.7	Spindle retaining disc in place												
6.8	Indicator plates in place												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

WATER CHECKLIST W6
PROPERTY SERVICE INSTALLATION

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from: to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
7.1	Plumber holding relevant accreditation on site							AS/NZS3500					
	Property boundaries located							AS/NZS3500					
7.2	Install property service (main to meter) - Drinking Water							AS/NZS3500					
7.3	Install property service (main to meter) - Recycled Water							AS/NZS3500					
7.4	Drinking water riser is plain copper							AS/NZS3500					
7.5	Recycled water riser is coloured lilac							AS/NZS3500					
7.6	Drinking water riser has correct ball valve							AS/NZS3500					
7.7	Recycled water riser has correct ball valve							AS/NZS3500					
7.8	Risers are a minimum 300mm apart							Meter Fit Policy					
7.9	Drinking Water non-return valve fitted												
7.10	Termination of property service as per standard							AS/NZS3500					
7.11	Embedment and trench fill – drinking water property services							AS/NZS3500					
7.12	Embedment and trench fill – recycled water property services							AS/NZS3500					
7.13	Clearances from other services							AS/NZS3500					
7.14	Plumbers Compliance Certificate supplied drinking water property service/s							AS/NZS3500					
7.15	Plumbers Compliance Certificate supplied for recycled water property service/s							AS/NZS3500					
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

WATER CHECKLIST W7
PRE-CONNECTION INSPECTION

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:										CONSULTING ENGINEER:											
PIPE TYPE:										CLASS:											
CONTRACTOR:																					
Date from:		to:		SITE						DATE SITE 1:		SITE 2:		SITE 3:		SITE 4:		SITE 5:		SITE 6:	
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD				COMMENT				Signatures					
8.1	WAC compiled																				
8.2	Compaction and concrete tests																				
8.3	Pressure test results											COUNCIL HOLD POINT									
8.4	Marking tape in place & tested																				
8.5	Surface boxes and surrounds level																				
8.6	Indicator plates in place																				
8.7	Fitting box lids positioned correctly																				
8.8	Hydrant lugs positioned correctly																				
8.9	Fitting bolts protected to Standard																				
8.10	Min 100mm max 200mm to hydrant lugs																				
8.11	Shroud assembly to Standard																				
8.12	Min 100mm max 350mm to top spindle																				
8.13	Extension spindle in place (if req.)																				
8.14	Spindle retaining disc in place (if req.)																				
8.15	Chambers sized to Standards																				
8.16	Chamber ladder or step irons to Standards																				
8.17	Chamber drainage adequate & to Standards																				
8.18	Scour outlet protected from erosion																				
8.19	Site restored satisfactorily																				
8.20	Chemical Analysis report																				
VARIATIONS AND CHANGES:										SITE INSTRUCTIONS:											
COMMENT:																					

APPENDIX D – Pump Station Inspection and Test Plan Template

INSPECTION AND TEST PLAN – WATER/SEWAGE PUMP STATION

Developer:		Consultant Engineer:		Consultant Engineer Representative:				
Project:		Contractor:		Contractor Site Representative:				
Description:		Sub-contractor:		Witness, Hold & Surveillance points added to ITP				
		Field Tester:						
Location:		ITP Prepared by:		Reviewed by:		Council Representative		
		Date / /		Date / /				
No	Construction/Inspection Activity	Inspection Procedure & Acceptance Criteria		Contractor	Consult. Engineer	Council *	Record	Comment
1	Pre-start/Site establish	Pre-Start Meeting Checklist. Site establishment visual check. Checklist completed. (PS1)		I	H	H	Checklist PS1	
2	Approved materials on Site/delivered	Visual check approved materials. Quantity and condition. Checklist completed (PS2)		I	I	S	Checklist PS2	
3	Excavation	Visual inspection to FNQROC Standards. Checklist completed. (PS3)		I	I	S	Checklist PS3	
4	Foundations	Visual and dimensional check to FNQROC Standards.		I	W	W	Checklist PS4	
5	Base slab	Visual inspection to FNQROC Standards.		I	H	I	Checklist PS4	
6	Reinforcement and formwork	Visual inspection to FNQROC Standards.		I	H	W	Checklist PS4	
7	Anchor/Thrust Blocks	Visual and dimensional check to FNQROC Standards.		I	H	H		
8	Embedment and Backfill	Visual check and compaction to FNQROC Standards		I	H	H	Compaction test results	
9	Electrical/Scada	Review certification and visually check installation to FNQROC standards.		I	W	I	Certification	
10	Lifting Chain	Review certification.		I	I	I	Certification	
11	Surface fittings	Visual and dimension check to FNQROC Standards. Checklist completed (PS6)		I	I	S	Checklist PS5	
12	Disinfection	Disinfection to FNQROC Standards		I	H	H	Test Results	
13	Testing	Pressure test and Compaction test to FNQROC Standards		I	H	H	Test Results	
14	Pre-connection inspection	Visual inspection to FNQROC Standards. Checklist completed (PS6) Isolation procedure as per Job Specific Letter		H	H	H	Checklist PS6	
15	Commissioning of System	Visual and dimensional check to FNQROC Standards and, where required, removal of RPZD.		I	H	H	PS Commiss. Checklist	



CONSTRUCTION PROCEDURES - Appendices

Symbol	Legend	No	Amendment	Date	Reviewed	Validation
I	Inspection					I certify that the works have been constructed in accordance with FNQROC Standards and the Inspection and Test Plan Consulting Engineer Date / /
H	Mandatory Hold Point					
W	Witness Point					
S	Surveillance					

* Council reserves the right to vary these requirements at any time

** Council's written approval **MUST** be obtained prior to varying these requirements

PUMP STATION CHECKLIST PS1 PRE-START AND SITE ESTABLISHMENT													
										* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL			
PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
1.1	Plan current and on site												
1.2	Pre construct report inc. photographs												
1.3	Property Entry Agreement												
1.4	Road opening requirements												
	Fees paid												
	Traffic mgt plan implemented												
1.5	Environmental Management Plan on site and implemented												
1.6	Wh&S Plan on site and implemented												
1.7	Receiving sewer located												
1.8	Specification on site												
1.9	Footways to finished levels												
1.10	Survey pegs in place							Surveyor					
1.11	Job set out												
1.12	All services located							‘Dial Before You Dig’, services search and Relevant Authorities					
1.13	All services marked												
1.14	Contractors holding relevant accreditation on site												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													



PUMP STATION CHECKLIST PS2 APPROVED MATERIALS ON SITE AND DELIVERED													
* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL													
PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
2.1	Delivery Inspection												
2.2	Types and sizes to current plan												
2.3	Marking tape												
2.4	Bedding material												
2.5	Trench fill												
2.6	Fittings												
2.7	Surface Fittings												
2.8	Pre Cast chambers												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													



PUMP STATION CHECKLIST PS3 – PAGE 1 OF 2 EXCAVATION										
PROJECT:								CONSULTING ENGINEER:		
Date from: to:								CONTRACTOR:		
PIPE TYPE:								CLASS:		
DAY								DATE DAY 1: DAY 2: DAY 3: DAY 4: DAY 5: DAY 6:		
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT	Signatures
4.1	Environmental Management Plan on site and implemented									
4.2	Traffic Management Plan on site and implemented									
4.3	Services exposed									
4.4	Clearance from Services									
4.5	Trench width mm									
4.6	Trench depth mm									
4.7	Trench shoring									
4.8	Excavation prior to placement of backfill								COUNCIL HOLD POINT	
4.9	Embedment									
	Compaction									
	Bedding									
	Surround									
	Overlay									
	Testing									
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:		
COMMENT:										

**PUMP STATION CHECKLIST PS3 – PAGE 2 OF 2**
EXCAVATION AND PIPE LAYING* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
Date from:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
to:		DAY						DATE DAY 1:	DAY 2:	DAY 3:	DAY 4:	DAY 5:	DAY 6
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT			Signatures	
4.10	Valves, Hydrants & Surface fittings installed												
	Shroud assembly												
	Valve anchorage												
4.11	Marking tape												
	Correct location												
	Connected to fittings												
4.12	Concrete												
	Trench stops in place												
	Bulkheads in place												
	Thrust blocks in place								COUNCIL HOLD POINT				
	Embedment & Encasement in place												
4.13	Trench fill												
	Material												
	Compaction												
	Compaction Testing							NATA Certified Lab					
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													



PUMP STATION CHECKLIST PS4 – PAGE 1 OF 2 CHAMBERS													
* CLAUSE AND DRAWING NO. REFER TO FNQROC DEVELOPMENT MANUAL													
PROJECT:								CONSULTING ENGINEER:					
Date from:		to:		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:			
				CHAMBER				DATE CH 1:	CH 2:	CH 3:	CH 4:	CH 5:	CH 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD		COMMENT		Signatures	
	Finished surface levels supplied												
5.1	Base												
	Placement												
	Channels												
	First shaft section												
5.2	In-Situ Chamber												
	Formwork – correct sizing												
	Formwork – correct levels												
	Reinforcement												
	Cover												
	Concrete type to Specification												
	Step iron location and spacing												
	Dimension check												
	Cover and frame												
	Conduits												
5.3	Pre Cast Chamber												
	Shaft assembled in correct order												
	Step iron location and spacing												
	Sealing												
	Offset cone located correctly												
	Minimum one make up ring												
	Cover and frame												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

PUMP STATION CHECKLIST PS4 – PAGE 2 OF 2 CHAMBERS									
5.4	Ladders / handrails / step irons								
5.5	Sealing							Manufacturer Specification	
5.6	Drainage								
5.7	Security Grate lid								
	Plastering/rendering								
	Benching								
5.8	Operational access								
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:	
COMMENT:									



PUMP STATION CHECKLIST PS5
SURFACE FITTINGS

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
6.1	Surface boxes and surrounds to finished levels												
6.2	Surface box lids hinged in direction of traffic flow												
6.3	Shroud pipes assembled to Standards												
6.4	Fitting bolts protected to Standards												
6.5	Correct depth to Spindle tops												
6.6	Correct depth to Hydrant lugs												
6.7	Spindle retaining disc in place												
6.8	Indicator plates in place												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

**PUMP STATION CHECKLIST PS6
PRE-CONNECTION INSPECTION**

* CLAUSE AND DRAWING NO. REFER TO
FNQROC DEVELOPMENT MANUAL

PROJECT:								CONSULTING ENGINEER:					
		PIPE TYPE:		SIZE:		CLASS:		CONTRACTOR:					
Date from: to:		SITE						DATE SITE 1:	SITE 2:	SITE 3:	SITE 4:	SITE 5:	SITE 6:
ITEM	DESCRIPTION	1	2	3	4	5	6	MINIMUM STANDARD	COMMENT		Signatures		
7.1	WAC compiled												
7.2	Compaction and concrete tests												
7.3	Vacuum/air test results								COUNCIL HOLD POINT				
	Deflection test results												
	CCTV Inspection												
7.4	Marking tape in place & tested												
7.5	Surface boxes and surrounds level												
7.6	Indicator plates in place												
7.7	Chambers sized to Standards												
7.8	Chamber ladder or step irons to Standards												
7.9	Chamber drainage adequate & to Standards												
	Benching to Standard												
	Sealing to Standard												
7.10	Scour outlet protected from erosion												
7.11	Site restored satisfactorily												
VARIATIONS AND CHANGES:								SITE INSTRUCTIONS:					
COMMENT:													

APPENDIX E – Security Lodgement Form

Council

(INSERT COUNCIL NAME)

This sheet must be completed prior to the acceptance of any bond by Council.

Development Name:

Stage: File No.:

Applicant:

Consultant:

Purpose of Bond:

☐ Construction Security ☐ Uncompleted Works ☐ Defects Liability

.....

.....

Uncompleted Works Bond Assessment

Estimated time to complete bond works (not greater than 90 days)days

Current Contract Completion date

Anticipated Completion date

Consulting Engineer's estimated value of uncompleted works \$......

Bond Value (apply Factor 1.50) \$......

Construction/Defects Liability Bond Assessment

Consulting Engineer's estimated value of completed works \$......

Construction/Maintenance Bond Value (apply Factor 0.05)(min \$1,000.00) \$......

Council shall retain any interest accrued on cash monies paid to Council and held in trust fund by Council..

Consulting Engineer:

Signature: RPEQ No.:

Date:

APPENDIX F - Inspection Certificate for Witness/Hold Points

Council
(INSERT COUNCIL NAME)

This certificate registers evidence that the works as noted herein have been inspected by the Council officer noted below and were found to be satisfactory.

Development Name: File No.....

Development Location:

Consulting Engineer:

Contractor:

Works being Inspected / Tested / Witnessed:

.....
.....
.....
.....

Defaults/Corrective Action Required

.....
.....
.....
.....

Defaults Corrected: Y N N/A

Council Inspector Signature

Name of Inspector

Date of Inspection:

APPENDIX G – Works Acceptance Inspection Checklist

COUNCIL:

DEVELOPMENT NAME: File No:

DEVELOPMENT

LOCATION:

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
ALLOTMENT DRAINAGE		
The works have been finally inspected and:		
a) Concrete catch drains constructed in approved location and to a satisfactory standard.		
b) Sediment basin, detention basins constructed in approved location and to a satisfactory standard		
c) Field Inlets constructed in approved location and to a satisfactory standard.		
d) Overland flow path constructed to correct profile.		
e) Inter allotment catch bunds constructed to correct profile		
f) Pipework has been visually inspected and is satisfactory ie., – alignment and grade – free of debris and siltation – no visual sign of trench subsidence – outlets are satisfactory		
g) Lots not provided with Allotment Drainage can be drained to the kerb and channel.		
STORMWATER DRAINAGE SYSTEM		
The works have been finally inspected and:		
a) Pipe layout is as per plan or approved amendments with respect to pipe size, levels and location.		
b) Pipework has been visually inspected and is satisfactory, ie., – alignment and grade – free of debris and siltation – pipe joints satisfactory – lifting plug holes sealed – no visible sign of trench subsidence – no damaged pipes		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
c) Gully pits and manholes have been constructed to the correct standards, ie., – correct type of grate or cover – lintels – side entry slots – benching (no water ponding) – grates are satisfactorily seated in frames – weepholes provided to bedding material – no damaged structures – grouted/mortared joints – converter slabs/sections mortar bedded – correct drops through gullies/manholes – all lids/grates finished to match surface level		
d) All density tests to backfill are available and satisfactory.		
e) Material gradings are available for bedding material and satisfactory.		
f) Outlet/Inlet structures are satisfactorily constructed and are free from scour or siltation.		
g) All manhole and gully pit pipe connections are mortared flush with the walls and no pipe reinforcement is exposed.		
h) Roadside table drains have been finally inspected are satisfactory, ie., – cut to design profiles – lining of channel is to the required thickness and reinforcement, with appropriate weepholes – bank stabilised and weed free		
i) Overland flow, the works have been finally inspected and appropriate flow paths are provided and clear of obstruction.		
j) Outlets and outfalls have been constructed to control discharge flow in accordance with the plans.		
k) Subsoil drainage discharges to gullies or other approved points of discharge.		
l) All grassing requirements to channels, swales, outlets, inlets etc have been completed.		
m) CCTV inspections of stormwater pipes ^{TRC}		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
WATER QUALITY The works have been finally inspected and:		
a) Water Quality structures have been constructed in accordance with approved engineering drawings		
b) Structures are free of debris and sediment		
EROSION AND SEDIMENT CONTROL The works have been finally inspected and:		
a) Control structures required until the site is stabilised in accordance with the Contractor's ESCP are in place. All exposed surfaces are treated/stabilised		
b) ESC on site reflects plan on file		
c) Structures (sediment basins, silt fencing, silt socks) are free of debris and sediment		
EARTHWORKS The works have been finally inspected and:		
a) Toe of batters not on Council road reserve except as approved.		
b) Retaining walls clear of road reserve except as approved.		
c) Retaining walls constructed in accordance with drawings.		
d) Batter slopes constructed in accordance with drawings.		
e) Batter slopes stabilised against erosion.		
f) Interim drainage constructed in accordance with drawings.		
g) All areas disturbed by the works have been rehabilitated.		
h) Allotment levels are as per the design plans		
i) Verge levels are as per the design plans		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
SEWER RETICULATION The works have been finally inspected and:		
a) Pipe layout is as per the plan or approved amendments with respect to pipe size, levels, and location.		
b) Pipework has been visually inspected and is satisfactory, ie: – pipework flush with internal walls of manhole – alignment and grade – flexible joints – line flushed and clean – no visible sign of trench subsidence – A density test of backfill are available and satisfactory. – CCTV survey results submitted and satisfactory – Spreadsheet of CCTV asset management data and SER/STR Score)		
c) Manholes and Maintenance Shafts have been constructed to the correct standards, ie., – cast in situ – benching – curvature satisfactory – no ponding – profile satisfactory – no weeps (free of infiltration) – concrete work – no honey combing – covers – covers checked to be gas tight – correct type – imprint in accordance with standards – depth of cover surround – depth of top slab – location – relative to lot boundaries – 50-75mm proud of finished surface level.		
d) Material gradings for bedding material are available and satisfactory.		
e) Pressure test results are available and satisfactory.		
f) Manhole hydrostatic test all satisfactory.		
g) All property connection branch markers located.		
h) Sewerage connection Private Works fees paid		
i) On site Sewer Report Provided (if applicable)		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
j) PUMP STATION – refer separate PS Checklist - Pre-commission with generator - Full commission with connected power		
WATER RETICULATION The works have been finally inspected and:		
a) Pipe layout and services fixtures (valves and hydrants) is as per the plan or approved amendments with respect to pipe size and location.		
b) Pipework has been pressure tested in accordance with Council's requirements and test results are available and satisfactory.		
c) Pipework has been chlorinated in accordance with Council's requirements.		
d) There are no visible signs of trench subsidence or leaks.		
e) Valves and hydrants have been inspected and are satisfactory, ie., - location - sets and surrounds correctly installed to prevent ingress of soil, etc. - mortar packing to boxes correctly completed - depth to top of hydrant or valve stem within limits - dust caps to hydrants - colour of marker plate correct - direction of flow indicated - marking plates correctly installed - size of plate correct		
f) Material gradings for bedding material are available and satisfactory.		
g) Water supply connection Private works fees paid.		
h) PUMP STATION – refer separate checklist		
ROAD PAVEMENTS The works have been finally inspected and:		
a) Plan layout and geometry of road system is in accordance with the drawings.		
b) Finished levels at crown and channel are at design levels.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
c) Crossfalls are to the approved plan.		
d) AC is satisfactory with regard to finish and thickness.		
e) Joints in the seal (especially where various development stages apply) are flush.		
f) The sealed surface is free of blemishes.		
g) All compaction test, material quality (CBR), material grading, AC core tests are satisfactory and available.		
h) Ponding of stormwater does not occur.		
SEGMENTAL PAVERS (Where constructed) The works have been finally inspected and:		
a) All pavers have been correctly laid to pattern, within allowable tolerance, compacted, and the joints filled.		
b) Bedding sand for pavers drains to subsoil drainage.		
c) Pavers adjacent to concrete kerb and channel, edge restraints etc have been cut and laid in accordance with all relevant requirements.		
CONCRETE WORKS The works have been finally inspected and:		
a) The correct type has been used to all locations in accordance with drawings.		
b) Ponding of stormwater does not occur.		
c) Transitions and connection to existing construction are smooth and to a satisfactory standard of workmanship.		
d) Service conduit markers have been placed to kerb face.		
e) Lip and back of kerb are flush with the roadway and footpath respectively.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
f) All channelisation works and medians have been satisfactorily completed.		
g) Infill treatment of medians has been inspected and found satisfactory. Any landscaping has been completed to standard.		
h) Driveways to battle axe blocks are as per approvals in place		
i) Bin hardstands are as per approvals in place		
j) Subsoil drains have been provided (including under medians).		
k) Appropriate expansion and contraction joints provided		
l) Subsurface finish is to approved design and within tolerances		
FOOTPATHS The works have been finally inspected and:		
a) Profiles are as per plan.		
b) Footpath has been topsoiled and satisfactory.		
c) Footpaths have been stabilised/fully turfed.		
d) All service fixtures (such as valves etc.) flush with the surrounding footpath.		
e) Concrete footpaths have been constructed to Council specification – broom finish		
f) Kerb ramps constructed as required.		
g) Footpaths to be free of broken edges, lifting, rock and loose stones.		
BIKEWAYS The works have been finally inspected and:		
a) Location and width are as per the drawings.		
b) Kerb ramps and crossings are constructed.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
c) Safety rails and signs have been installed where required.		
LIGHTING / ELECTRICAL The works have been finally inspected and:		
a) Lighting has been installed and is operating as per approved design		
b) If lighting is yet to be installed, or made operational, copy of service agreement has been provided from the lighting/energy provider and all uncompleted works have been adequately guarded.		
c) Pad mount is installed in accordance with approved plans		
d) Overhead electrical infrastructure made safe until removal if in lot		
FENCING AND FEATURES The works have been finally inspected and:		
a) All fences including approved entrance features have been constructed within allotments. Survey pegs are visible.		
b) Specifically approved entrance features are constructed in accordance with the drawings.		
c) Entrance features and fences have satisfied Building Approvals (if required).		
d) Sound attenuation fences and/or mounds are constructed on private property and in accordance with the drawings where required.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
BUILDING/STRUCTURE		
The works have been finally inspected and:		
a) Council approval for building/structure		
b) retaining walls >1.0m RPEQ certified		
OTHER		
a) Approvals for completed works received from applicable referral agencies		
b) Traffic Management in place where required		
c) Street name signs, traffic signs and pavement marking have been installed.		
d) Works have not resulted in problems on neighbouring properties. Clearance letters from property owners are available where applicable.		
e) All boundaries of Subdivision/Development have been inspected to ensure works as constructed will not affect adjoining properties.		
f) All necessary testing to ensure the quality of the work has been carried out and results are available.		
g) Consulting Engineer's compliance certificate is completed (refer AP1 – Appendix A)		
h) "As Constructed" submission has been provided to Council and is to Councils satisfaction		
i) All allotment boundaries, easements etc, have been pegged.		
j) All test results and records have been compiled and stored in the Record Storage facilities of the Consulting Engineer's office and a copy forwarded to Council.		
k) All operating Manuals, maintenance procedures, mechanical warranties etc have been submitted to Council.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
l) Parkland is in a mowable condition where practical and free of rock and loose stones.		
m) Irrigation systems have been provided, are operating as designed and AS CON drawings provided.		
INSPECTOR'S SIGNATURE: NAME: SIGNATURE:DATE:...../...../..... CONSULTING ENGINEER: RPEQ No:DATE:...../...../.....		

APPENDIX H - Final Acceptance Inspection Checklist

COUNCIL:

DEVELOPMENT NAME: File No.....

DEVELOPMENT LOCATION:

ITEM	VERIFICATION	COMMENT
	(Yes/No/NA)	
STORMWATER DRAINAGE SYSTEM		
a) Pipework has been visually inspected and is satisfactory, ie.,		
- free of debris and siltation		
- pipe joints satisfactory with no deflection or movement		
- no visible sign of trench subsidence		
- no exposed reinforcing steel to cut pipe ends		
b) Gully pits and manholes have been visually inspected and are satisfactory, ie.,		
- no ponding		
- no excessive cracking or distress of reinforced concrete works		
- clear of silt and debris		
- all mortar is in place, no excessive spalling, flaking or cracking		
- no visible sign of subsidence		
c) overland flow paths clear		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
WATER QUALITY		
a) Water Quality Structures have been visually inspected and are satisfactory, ie.,		
- free of debris and siltation		
- no cracking or distress of concrete at fixing points		
- fasteners are secure		
- structures have not misaligned due to excessive loads		
- no corrosion at any location evident		
ALLOTMENT DRAINAGE		
a) Concrete catch drains have been visually inspected and are satisfactory, ie.,		
- clear of silt and debris		
- no damage or cracking		
- overland flow path profile maintained		
EARTHWORKS/SITE WORKS		
a) All batter slopes stable and no distress exhibited.		
EROSION AND SEDIMENT CONTROL		
a) Site has been visually inspected and has no obvious signs of erosion or sediment deposits and has achieved 80% grass cover		
b) Erosion and sediment control measures no longer required have been removed and rehabilitation works completed.		
c) Erosion and sediment control measures that remain are clean and operational		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
WATER RETICULATION		
a) No visible signs of trench subsidence.		
b) valves and hydrants have been inspected and are satisfactory, ie.,		
- no leaks		
- valve and hydrant markings		
- no damage		
SEWERAGE RETICULATION		
a) No visible signs of trench subsidence.		
b) Pipework has been visibly inspected and is satisfactory, ie.,		
- alignment satisfactory		
- clear of silt and debris (flushed)		
- no ponding		
- pipework not oval or compressed		
c) Manholes/structures have been visually inspected and are satisfactory, ie.,		
- benching no signs of cracking, spalling ok		
- no weeping or infiltration		
- no ponding or disposition of solids.		
d) PCB markers have been visually located		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
ROADWORKS		
a) Road pavement has been visually inspected and satisfactory, ie.,		
- no damage to Wearing Course		
- no ponding		
- clear of siltation and debris		
b) Kerb and channel has been visually inspected and is satisfactory, ie.,		
- no excessive cracking or distress to concrete works		
- no ponding		
- service conduit markers installed, painted satisfactorily		
- no differential settlement or dislocation of pavement surface and concrete kerb and channel.		
c) Linemarking and road signage satisfactory.		
BUILDING/STRUCTURE		
a) Building/Structure Inspected		
LANDSCAPING		
a) Landscaping has been provided for a minimum 13 week period prior to calling for the final works acceptance inspection.		
b) Landscaping assets are in an acceptable condition in accordance with the approved plan		
c) Existing trees have been protected in accordance with the approved plan.		

ITEM	VERIFICATION (Yes/No/NA)	COMMENT
PLAYGROUND / EMBELLISHMENTS		
d) Manufacturers/installers certification of installation		
e) Operation/maintenance guides and tools handed to council		
f) Park signs installed where required		
MISCELLANEOUS		
a) Footpaths and concrete works satisfactory. (no lifted/cracked paths)		
b) Bikeways and associated works satisfactory. (no lifted/cracked paths)		
c) Street name signage satisfactory.		
d) Alternative pavement surfacing (eg pavers, stamped concrete) is satisfactory		
e) Street lighting has been installed and is operating as per the approved design.		
f) All redundant overhead lighting/electrical in lots removed.		
OTHER MATTERS		
a) Specific matters in relation to the site.		
INSPECTORS SIGNATURE:		
NAME:		
CONSULTING ENGINEER:		
NAME:		
RPEQ:		
DATE:		

APPENDIX I - Sewerage and Water Pump Station Commissioning Checklists ^{CRC}

PRE-COMMISSIONING CHECKLIST FOR NEW ASSETS WATER SITES

REQUIREMENTS BY CONTRACTOR

The following checklist is required to be fully completed, signed and returned to Council before a joint commissioning is considered. Please note: in the event of a commissioning being abandoned due to works not completed or operational, Council will recover costs incurred.

SITE NAME RESERVOIR: _____ SITE NAME P/S: _____

- | | | |
|--|------------------------------|-----------------------------|
| • Ergon Power available | Yes ☐ | No ☐ |
| • Provision of sufficient water for all testing purposes | Yes ☐ | No ☐ |
| • Fresh water discharge flushing system operating as per design | Yes ☐ | No ☐ |
| • Ancillaries (GPO's, lighting etc) tested and working correctly | Yes ☐ | No ☐ |
| • Pre commission switchboard test completed | Yes ☐ | No ☐ |
| • Station telemetry points list supplied | Yes ☐ | No ☐ |
| • Established telemetry communications and verified inputs locally | Yes ☐ | No ☐ |

VERIFICATION OF AS CONSTRUCTED LEVELS AND OPERATION.

Note: Levels taken from floor level

Setting/Alarm	Measurement	Operational function (contractor to verify correct operation)	
Overflow level	m		
High Level alarm	m	Yes ☐	No ☐
Fill stop level	m	Yes ☐	No ☐
Fill request level	m	Yes ☐	No ☐
Low level alarm	m	Yes ☐	No ☐
Very Low Level Alarm	m	Yes ☐	No ☐
Level settings according to design	Yes	No	NA

Commissioning of Civil Works

Item	Complete	Date / Initials	Council Audit
Management			
1. Verify that consultant has provided all documentation. (as constructed details, operating manuals, process and instrumentation diagram, test results etc)	☐		
2. Consultants have supplied O&M manuals – incorporating all drawing both electrical & mechanical.	☐		
3. Consultants have supplied test results including performance pump curves	☐		
If NO then close audit.	☐		
Concrete			
1. Verify that there is no damage to any exposed concrete surface.	☐		
2. Verify that drainage of the site is away from the structures.	☐		
3. Verify that the below ground concrete structures are dimensionally correct and in accordance with the design drawings.	☐		
4. Verify no seepage through the concrete structure.	☐		
5. Verify the verticality of the structure is within tolerance in accordance with SEWL specifications.	☐		
6. Verify that all chamfers are provided in accordance with the design drawings.	☐		
7. Verify that the specified coating to the internal walls has been applied in accordance with the FNQROC Development Manual.	☐		
8. Thickness of internal coating tested (µm)	☐		

OH&S			
1.	Verify that ladder access meets OH&S requirements.	☐	
2.	Verify that access to valve chamber meets OH&S requirements.	☐	
3.	Verify that all ladders are provided with the extension above the FSL.	☐	
4.	Verify that the ladders have non-slip treads.	☐	
5.	Have safety cages been specified in accordance with Australian Standards.	☐	
6.	Verify that safety cages have been installed in accordance with the design drawings.	☐	
7.	Verify that adequate set down areas for the covers has been provided in accordance with OH&S requirements.	☐	
8.	Verify that no overhead cables restrict access via crane trucks.	☐	
9.	Verify Full Risk Assessment has been performed and permanent operational/ maintenance risks identified	☐	
10.	Verify that all signage pertaining to those risks is in place	☐	
11.	Facility Name and contacts signage	☐	
12.	Standby Emergency Generator – cover, refueling, manuals	☐	
13.	Adequate access, maneuver and parking for maintenance vehicles	☐	
14.	Switchboard access and clearance meets requirements under Electrical Safety Act	☐	
15.	Personnel/maintenance access ring install in side wall of reservoir	☐	

Products & Materials	Complete	Date/ Initials	Council Audit
1. Verify that all products incorporated on the project are included on the list of approved products for council.	☐		
2. Verify that all markings as required by council specification are visible on the covers.	☐		
3. Verify that the covers and frames are greased in accordance with the manufacturer's requirements.	☐		
4. Verify that the interchangeable multi part covers have lifting lugs on the beams for removal and covers have clockwise lifting key holes.	☐		
5. Verify that internal ladders are stainless steel.	☐		
Pipes and Fittings			
1. Verify that the valves are anticlockwise closing.	☐		
2. Verify that adequate supports have been provided for the valves.	☐		
3. Are valve extension spindles required?	☐		
4. Verify that there is sufficient clearance for maintenance.	☐		
5. Are adequate supports provided in accordance with standard drawings.	☐ /NA		
6. Verify that DICL pipe work has been provided to correct nominal diameter DN.	☐		
7. Verify that adequate supports for vertical pipe work has been provided in accordance with FNQROC Development Manual.	☐		
8. Verify that all gate valves operate through the full range and are left in the open position.	☐		
9. Verify that a flap valve has been installed on the valve chamber drain if required by the design.	☐		

10.	Verify that the specified coating for all valves has been applied.	☐		
11.	Verify that the specified coating for the pipe work has been applied.	☐		
12.	Verify that the specified bolting system on the flanges has been used.	☐		
13.	Verify that all valves can be removed through the available cover opening.	☐		
14.	Verify that a flanged dismantling joint has been provided to allow ease of removal of valves in accordance with standard drawings.	☐		
15.	Has the consultant submitted completed project ITP containing all signatures.	☐		
16.	Verify all steel metalwork has been Hot Dipped Galvanized.	☐		
17.	Verify all bolts are 316 stainless steel with 308 nuts.	☐		

Other Services.		Complete	Date / Initials	Council Audit
1.	Verify that the water service has been fitted with an approved back flow prevention device.	☐		
2.	Verify that all conduits through the walls have been sealed.	☐		
3.	Verify that electricity is below ground, not above.	☐		
4.	Verify that the sealed access track is in accordance with the design drawings and the FNQROC Development Manual.	☐		
5.	Verify that adequate site drainage has been provided.	☐		
Restoration.				
Verify that the site restoration has been completed.		☐		

Testing.				
1.	Verify that Factory Acceptance Testing of the switchboard was completed	☐		
2	Verify that pump tests passed	☐		
3	Verify that rising main has been tested	☐		
Mechanical Equipment.				
1.	Refer to receiving Council for specific requirements	☐		

Electrical Equipment.				
1.	Verify identification plates has been fitted to the electrical cabinet.	☐		
2.	Verify that the telemetry antenna has adequate protection in accordance with specification.	☐		
3.	All wiring is installed, terminated and tagged as per drawings.	☐		
4.	All Earthing systems are installed & tested to Australian standards.	☐		
5.	Switchboard testing - Fault protection, circuit Breakers & cable insulation tests have been carried out. Results Attached.	☐		
6.	Verify that lighting within the switchboard has been provided.	☐		
7.	Verify that the Council locks fitted to switchboard and operational.	☐		
8.	Verify that quick link generator connectors provided if applicable.			

Security.	Complete	Date / Initials	Council Audit
1. Verify that the security fencing has been installed in accordance with the design drawings.	☐		
2. Verify that council keyed locks installed.	☐		
3. Verify that the switchboard meter cabinet has been fitted with Ergon Locks.	☐		
Have all NCR items been resolved (including any raised as a result of this audit)?	☐		
If YES Issue Acceptance of Works., and close audit.	☐		
Would outstanding NCR items impact on the ability to operate the pump?	☐		
If YES then close audit (wait for NCRs to be resolved).	☐		
Has Council authorised the pumps to remain on?	☐		

If YES, details of the person who authorised this.

NAME.....

SIGNATURE:.....

POSITION:.....

DATE:.....

APPENDIX I - Sewerage and Water Pump Station Commissioning Checklists

CRC

REQUIREMENTS BY CONTRACTOR

The following checklist is required to be fully completed, signed and returned to Council before a joint commissioning is considered. Please note: in the event of a commissioning being abandoned due to works not completed or operational, Council will recover costs incurred.

STATION NAME: _____ STATION NO: _____

- | | | |
|--|------------------------------|-----------------------------|
| • Ergon Power available | Yes ☐ | No ☐ |
| • Provision of sufficient water for all testing purposes | Yes ☐ | No ☐ |
| • Fresh water discharge flushing system operating as per design | Yes ☐ | No ☐ |
| • Ancillaries (GPO's, lighting etc) tested and working correctly | Yes ☐ | No ☐ |
| • Pre commission switchboard test completed | Yes ☐ | No ☐ |
| • Station telemetry points list supplied | Yes ☐ | No ☐ |
| • Established telemetry communications and verified inputs locally | Yes ☐ | No ☐ |

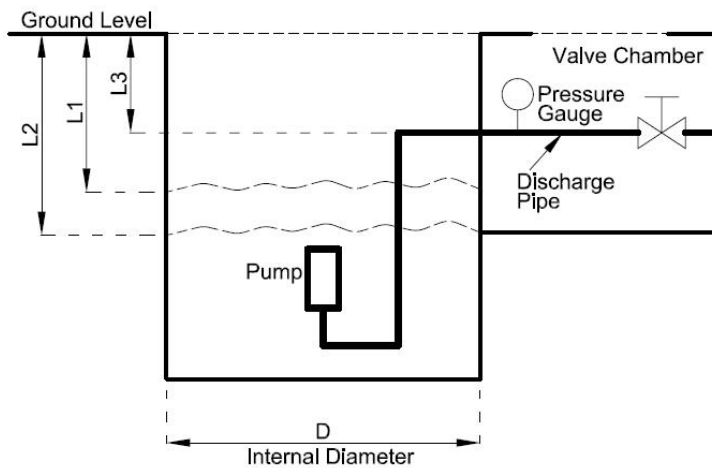
VERIFICATION OF AS CONSTRUCTED LEVELS AND OPERATION.

Note: Levels taken from top of well cover down

Setting/Alarm	Measurement	Operational function (contractor to verify correct operation)
Spill alarm	m	Yes ☐ No ☐
H/L back up pump 2 start	m	Yes ☐ No ☐
H/L back up pump 1 start	m	Yes ☐ No ☐
Probe high level alarm	m	Yes ☐ No ☐
H/L pump cut out/ Transfer	m	Yes ☐ No ☐
Duty pump cut-in	m	Yes ☐ No ☐
Duty pump cut out	m	Yes ☐ No ☐
Probe low level alarm (indication only)	m	Yes ☐ No ☐
Level settings according to design		Yes ☐ No ☐

PROCEDURE FOR PERFORMANCE TESTING OF PUMPS

Contractor to complete for all pumps :-



General Sewerage Pump Station Well

Date _____

Pump No _____

Pump _____ make/model:

Electric motor kw _____ SPEED _____ RPM

Serial no: = _____

Measure inside diameter of well:- $D =$ _____ m

Measure $L3 =$ _____ m

Pump duty head $H =$ _____ m

Pump duty flowrate $Q =$ _____ L / S

(A) **QUICK CHECK OF PUMP GENERAL CONDITION - HEAD TH1 (AT VALVE CLOSED)**

L1 , L2 - operating levels

Run pump for one minute with the delivery valve open.

Close the valve.

Read L1 in meters at the beginning of one minute.

Read L2 in meters at the end of this minute.

Read AMP meter.

Read pressure gauge (HD) in meters at the end of one minute.

Record :

L1

L1 = m

L2

L2 = m

Delivery Gauge reading HD

HD = m

Height Correction HC = L2 – L3

HC =m

Total head TH1 = HD + HC

TH1 =m

Check TH1 against the pump curve at no flow condition

Check AMPs on pump curve

(B) **QUICK CHECK OF INFLOW Q1 (IF INFLOW BLOCKED OFF - GO TO (C))**

L1 , L2 - operating levels

Switch off the pump and keep the delivery valve closed.

Read L1 in meters at the beginning of one minute.

Read L2 in meters at the end of this minute.

L1

L1 = m

L2

L2 = m

Calculate flowrate Q1 during this minute

Q1 = l/s

(C) **NORMAL FLOWRATE Q2 TEST (VALVE OPEN)**

L1 , L2 - operating levels

Run pump for one minute with the delivery valve open.

Read L1 in meters at the beginning of one minute.

Read L2 in meters at the end of this minute.

Read AMP meter.

Read HD in meters at the end of one minute.

Record :

(C)

L1

L1 = m

Cont.

L2

L2 = m

Delivery Gauge reading HD

HD = m

Height Correction HC = L2 – L3

HC =m

Total head TH2 = HD + HC

TH2 =m

Calculate flowrate Q2 during this minute

Q2 = l/s

Check AMPs on pump curve



(D) **QUICK CHECK OF INFLOW AGAIN Q2 (IF INFLOW BLOCKED OFF GO TO (E))**

L1 , L2 - operating levels
Switch off the pump and keep the delivery valve closed.
Read L1 in meters at the beginning of one minute.
Read L2 in meters at the end of this minute.

L1 L1 = m
L2 L2 = m

Calculate flowrate Q3 during this minute

Q3 = l/s

(E) **SUMMARY**

If inlet is blocked off for the time of the test the duty head H and duty flowrate Q should be similar to TH2 and Q2.

If inlet is not blocked off for the time of the test the duty head should be similar to TH2 and duty flowrate should be similar to :

$$Q_D = Q_2 + (Q_1 + Q_3)/2$$

End of testing of pumps

Pressure gauges calibrated according to QA requirements:- Yes No

Pump performance satisfactory:- Yes No

Test performed by :- _____

Pump performance results as compared to tender offer satisfactory? Yes No

General Contractor Comments:



CONSTRUCTION PROCEDURES - Appendices

All information has been supplied and verified ready for Commissioning.

Signed (Contractor)

Date

Print name

Note: The telemetry is to be fully ready for commissioning

COUNCIL use only

Level information and Operational function satisfactory	Yes	No
Pump performance results as compared to tender offer satisfactory	Yes	No
Pump motor information readings satisfactory	Yes	No
Telemetry commissioned satisfactorily	Yes	No

General Council Comments:

All information has been supplied and verified at Contractor Pre-commissioning

Signed (Council)

Date

Print name

Commissioning of Civil Works

Item	Complete	Date / Initials	Council Audit
Management			
1. Verify that consultant has provided all documentation. (as constructed details, operating manuals, Process and Instrumentation Diagram (P&ID), test results etc)			
If NO then close audit.			
Concrete			
1. Verify that the concrete slab is 300mm above the finished surface level.			
2. Verify that the pump station concrete works is as designed e.g. Cast in-situ.			
3. Verify that there is no damage to any exposed concrete surface.			
4. Verify that drainage of the site is away from the Pump Station.			
5. Verify that the surface dimensions of the top slab are in accordance with the design drawings.			
6. Verify that the below ground concrete structures are dimensionally correct and in accordance with the design drawings.			
7. Verify no seepage through the concrete structure.			
8. Verify the verticality of the structure is within tolerance in accordance with sewer specifications.			
9. Verify that all chamfers are provided in accordance with the design drawings.			
10. Verify that the pump well benching has been provided in accordance with design drawings.			
11. Verify that the specified coating to the internal walls has been applied in accordance with the FNQROC Development Manual.			
12. Thickness of internal coating tested (µm)			

Item	Complete	Date / Initials	Council Audit
OH&S			
1. Verify that ladder access to dry wells only, meets OH&S requirements.			
2. Verify that ladder access to valve chamber meets OH&S requirements.			
3. Verify that all ladders are provided with the extension above the FSL.			
4. Verify that the ladders have non-slip treads.			
5. Have safety cages been specified in accordance with Australian Standards.			
6. Verify that safety cages have been installed in accordance with the design drawings.			
7. Verify that adequate distance between wet well opening and switchboard is in accordance with OH&S requirements and the FNQROC Development Manual.			
8. Verify that adequate set down areas for the covers has been provided in accordance with OH&S requirements.			
9. Verify that no overhead cables restrict access via crane trucks.			
10. Verify Full Risk Assessment has been performed and permanent operational/ maintenance risks identified			
11. Verify that all signage pertaining to those risks is in place			
12. Facility Name and contacts signage			
13. Standby Emergency Generator – cover, refueling, manuals			
14. Adequate access, maneuver and parking for maintenance vehicles			
Products & Materials			
1. Verify that all products incorporated on the project are included on the list of approved products for Council.			

Item	Complete	Date / Initials	Council Audit
2. Verify that all markings as required by Council specification are visible on the covers.			
3. Verify that the covers and frames are greased in accordance with the manufacturer's requirements.			
4. Verify that the covers are gas tight			
5. Verify that the interchangeable multi part covers have lifting lugs on the beams for removal and covers have clockwise lifting key holes.			
6. Verify that council approved penstock stuffing box is installed.			
7. Wet well covers are to be in accordance with the FNQROC Development Manual			
8. Wet well covers are to be lockable			
Pipes and Fittings			
1. Verify that the valves are anticlockwise closing.			
2. Verify that there is only one inlet pipe to the pump station.			
3. Verify that pumps are clear of all inlet pipework.			
4. Verify that adequate supports have been provided for the valves.			
5. Are valve extension spindles required.(includes penstock).	/NA		
6. Verify that there is sufficient clearance for maintenance.			
7. Are adequate supports provided in accordance with Council standard drawings.	/NA		
8. Verify that DICL pipe work has been provided to correct nominal diameter DN.			

Item		Complete	Date / Initials	Council Audit
9.	Verify that adequate supports for vertical pipe work has been provided in accordance with FNQROC Development Manual (i.e. vibration not noticeable when pumps operating).			
10.	Verify that all gate valves operate through the full range and are left in the open position.			
11.	Verify that a flap valve has been installed on the valve chamber drain if required by the design e.g. when overflow levels higher than valve pit base.			
12.	Verify that bleeders have been installed on the NRV's and NRV's have counterweights.			
13.	Verify that probe stilling tube has been installed as per Council standard			
14.	Verify that the Council specified coating for all valves has been applied.			
15.	Verify that the Council specified coating for the pipe work has been applied.			
16.	Verify that the pipe work for the in coming sewer is in accordance with the design drawings including dropper pipe.			
17.	Verify that the specified bolting system on the flanges has been used.			
18.	Verify that all valves can be removed through the available cover opening.			
19.	Verify that a flanged dismantling joint has been provided to allow ease of removal of valves in accordance with standard drawings.			
20.	Has the consultant submitted completed project ITP containing all signatures.			
21.	Verify all steel metalwork has been Hot Dipped Galvanized.			
22.	Verify all bolts are 316 stainless steel with 308 nuts.			
23.	Verify that the emergency pump out pipework has been installed.			

Item	Complete	Date / Initials	Council Audit
24. Verify that either rising main air releases or scours are provided at the pump station.			
Other Services			
1. Verify that the water service has been fitted with an approved back flow prevention device.			
2. Verify that a 25mm water service has been provided.			
3. Verify that all conduits through the walls have been sealed to prevent odour escaping.			
4. Verify that electricity is below ground, not above.			
5. Verify that the sealed access track is in accordance with the design drawings and the FNQROC Development Manual.			
6. Verify that adequate site drainage has been provided.			
7. Verify that wet well washers have gate valves and regulators fitted if fitted.			
Restoration.			
Verify that the site restoration has been completed.			
Testing.			
1. Verify that pump well infiltration test passed.			
2. Verify that pump draw down tests passed.			
3. Verify rising main has been tested.			
4. If pump station has been constructed as open cut, verify compaction standard under pipes.			
5. Verify that Factory Acceptance Testing of the switchboard was completed			

Item	Complete	Date / Initials	Council Audit
Mechanical Equipment.			
1. Verify that the guide rails comply with the standard drawings.			
2. Verify that the lifting chain complies with Council specification.			
3. Verify that the Pump footstool has been secured to wet well floor with appropriate chemical anchors & have 316 S/S bolts.			
4. Are wet well washers specified on design drawings.			
5. Verify that they meet the Council specified requirement.			
6. Verify that pumps can be fully removed from pump station without disconnecting the guide rails.			
Electrical Equipment.			
1. Verify that the Station Identification plate has been fitted to the electrical cabinet.			
2. Verify that the telemetry antenna has adequate protection in accordance with Council specification.			
3. Verify that lighting within the switchboard has been provided.			
4. Verify that the Council locks fitted to switchboard and operational.			
5. Verify that quick link generator connectors provided.			
6. Verify that the Process and Instrumentation Diagram is in switchboard cabinet			
Security.			
1. Verify that the security fencing has been installed in accordance with the design drawings.			
2. Verify that Council keyed locks installed.			



CONSTRUCTION PROCEDURES - Appendices

Item	Complete	Date / Initials	Council Audit
3. Verify that the switchboard meter cabinet has been fitted with Ergon Locks.			
Have all NCR items been resolved (including any raised as a result of this audit)?			
If YES Issue Acceptance of Works., and close audit.			
Would outstanding NCR items impact on the ability to operate the pump?			
If YES then close audit (wait for NCRs to be resolved).			
Has Operations authorized the pumps to remain on?			
If YES record the name of the person who authorized this. NAME.....			
Lift Station			
1. Has discharge pit been lined in accordance with the FNQROC Development Manual			
2. Has benching been completed			
Overflow			
1. Verify overflow has been constructed to design drawings and FNQROC Development Manuals including levels.			
2. Verify flap valves are in place and operational.			
3. Verify covers are in place and are to class specified.			
4. Verify Overflow can be accessed for maintenance.			

APPENDIX J - Statement of Compliance “As Constructed”

Council

Name of Development:

Location of Development:

Applicant:

Consulting Engineer:

Surveyor:

It is hereby certified that the “As Constructed” drawings submitted herewith have been prepared, checked and amended in accordance with the requirements of the FNQROC Development Manual and that the completed works comply with the requirements therein.

Certification by Surveyor attached ☐ Yes

(Note: Certification is to be in accordance with the Development Manual).

Compliance with the manual Design Intent and Function not compromised by the “As Constructed” Works.	Compliance Yes / No	Non-Compliance refer to attached redesign of works to ensure satisfactory performance
Earthworks		
Roadworks		
Stormwater Drainage - Minor Flow System and Structures - Major Flow System and Structures		
Water Reticulation		
Sewerage Reticulation		
“As Constructed” Documentation		

Conscientiously believing the above statements to be true and correct:

Consulting Engineer

Name in Full **RPEQ No.**

Signature

Date



APPENDIX K - Example of Surveyor Certification of “As Constructed” Works

We _____ (Name of Surveying Consultant) hereby certify that the locations, surface and invert levels of all works and infrastructure presented on the drawings noted below and in the digital data have been surveyed and meet the accuracy standards as defined within the FNQROC Development Manual.

Surveyor / Director:

Date:

Drawings and Documents pertaining to the above:

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

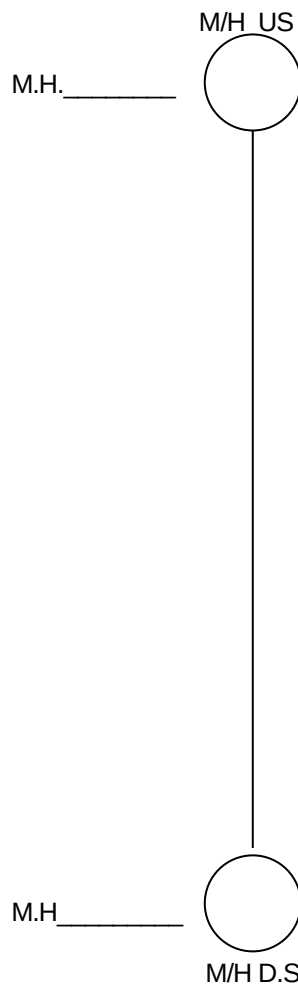


APPENDIX L As Constructed Data Sewer Property Connection Branches

Development Name:..... Date:.....

Contractor:.....By:.....

Stage:.....



Φ	D/S IL	U/S IL	Grade	Length

PROPERTY CONNECTIONS

	D/S Mh No.	U/S Mh No.	Total	Lot. No.
Ch				
IL				
SL				
Ch				
IL				
SL				
Ch				
IL				
SL				
Ch				
IL				
SL				
Ch				
IL				
SL				
Ch				
IL				
SL				
			U/S M/H	
			TOP RL	

NOTES:.....
.....

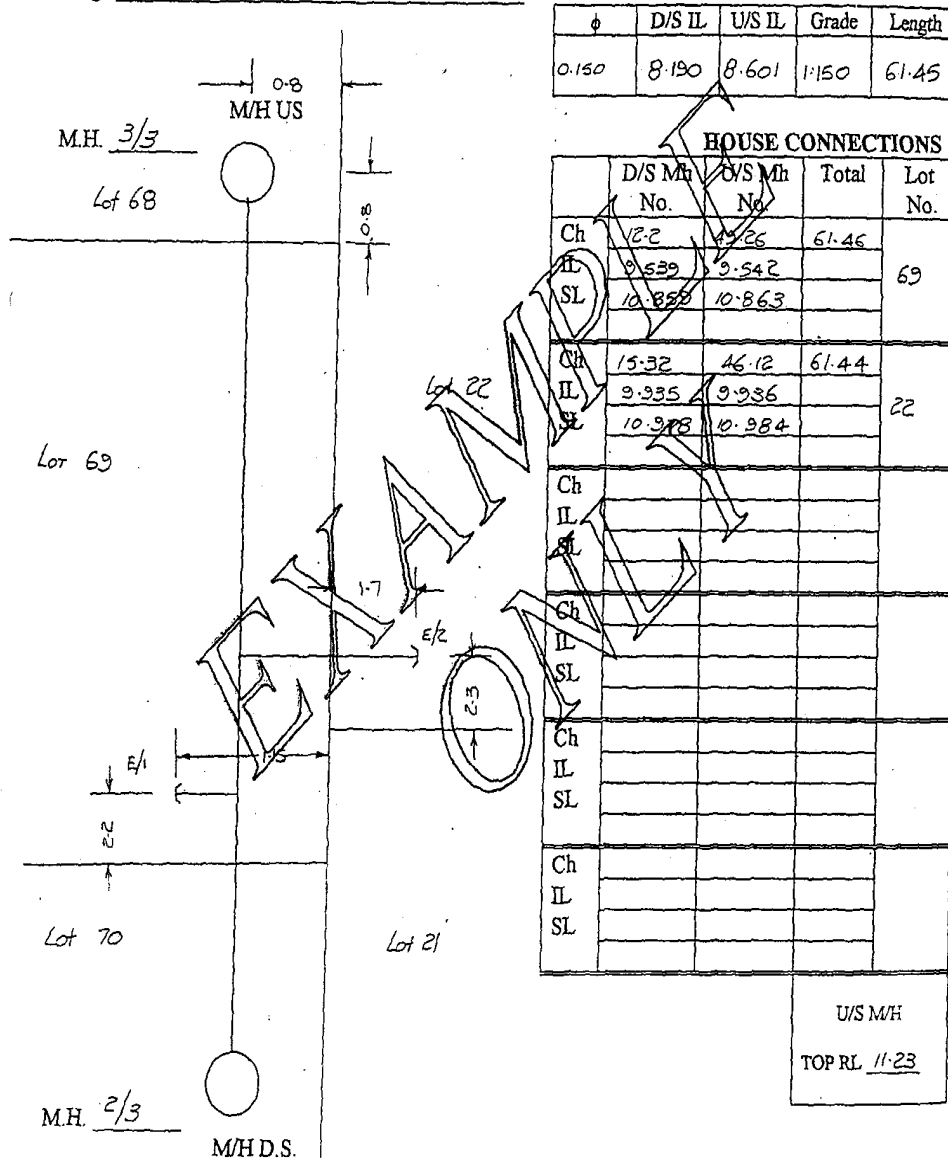
Certified as True and Correct location: Surveyor:

Signature:

Date:

AS-CONSTRUCTED DATA SEWER HOUSE CONNECTION BRANCHES

Development Name Stillson Estate Date 16-5-96
Contractor Digwell Constructions By KLM
Stage 2



NOTES:

Certified as True and Correct on Behalf of the Contractor: Name: John Treadwell

Signature: J Treadwell

Date: 18/5/96

APPENDIX M - As Constructed Data Submission Form

		AS CONSTRUCTED DATA SUBMISSION FORM For Consulting Engineers' or Surveyors' Submission Of As Constructed Data			
Applicant Details	Development Name (Title on approved Engineering Drawing)				
	File Number		Stage Number		
Property Address	Real Property Description				
	Address				
	Company Name				
	Address				
As Constructed Documentation Submitted			Hardcopy	Electronic Copy	Amendment # (A,B,C....)
	Survey Datum		☐	☐	
	Water		☐	☐	
	Sewer		☐	☐	
	Drainage		☐	☐	
	Road		☐	☐	
	Parks / Landscape / Structures		☐	☐	
	Utilities		☐	☐	
	Contours		☐	☐	
	Digital field survey data, ASCII		☐	☐	
Other Documentation Submitted		Previously Submitted	Attached	N/A	Amendment #
	Engineering drawings in electronic format (AUTOCAD):	☐	☐	☐	
	Landscaping and park embellishments in electronic format (AUTOCAD):	☐	☐	☐	
	Building / Structural Certification	☐	☐	☐	
	Subgrade CBR results	☐	☐	☐	
	Maintenance Manuals	☐	☐	☐	
Office Use Only	Date Received		DISK Number		
	Officer Checked		Document Number		



APPENDIX N - (Pending) Permanent Survey Mark Coordination Request Form

APPENDIX O - Deleted

APPENDIX P - “As Constructed” Digital Data and Drawing Standards

1. SURVEY DATUM

Survey Datum is the framework of all geospatial information and provides the starting reference that not only supports the accuracy and integrity of survey data, but also provides the geospatial correlation of assets with other data sets. The following publications, or their successors, are to be used in conjunction with this section:

Standards and Practices for Control Surveys (SP1) - Inter-governmental advisory Committee on Surveying and Mapping (ICSM) publication. Available in .PDF format from the ICSM website. <https://www.icsm.gov.au/sites/default/files/2017-05/sp1v1-7.pdf>

- GDA Technical Manual - Inter-governmental advisory Committee on Surveying and Mapping (ICSM) publication. Available in .PDF format from the ICSM website. <http://www.icsm.gov.au/datum/gda2020-and-gda94-technical-manuals>
- *Cadastral Survey Requirements* – Refer to the relevant State Government website e.g. in 2019 this is <https://www.dnrm.qld.gov.au/?a=105601> publication available in PDF format.

a. Datum Recorded

The datum adopted must be recorded, allowing full traceability back to the origin. As the Department of Natural Resource and Water carry out periodic adjustments on both the coordinate and level networks, information to be included as part of the “As Constructed” submission must include a copy of the basic elements of traceability that include but not restricted to:

- i. Starting point of Datum.
- ii. State the type, material and condition of marks used
- iii. Show all calculations relating to datum establishment i.e. copy of traverse and levelling details
- iv. Table of adjusted coordinates, coordinate system, datum and zone.
- v. Baseline closure details from processing software.
- vi. Report on adjustment (generated from software).
- vii. Network map (A3 .pdf).
- viii. Permanent Survey Mark Reports (pdf).

This information is necessary to allow subsequent re-computation of the datum's for any future corrections made to the network.

b. Co-ordinate Datum

Co-ordinate Datum may be:

- i. Assumed (arbitrary) plane co-ordinates – No previous co-ordinates and no correlation required. Used only for where work is undertaken in an already proclaimed survey area or for subdivisions of five (5) lots or less.
- ii. MGA co-ordinates – grid co-ordinates from the adjustment of a survey traverse to a minimum of two (2) but preferably three (3) or more appropriate survey marks with MGA coordinates. Appropriate survey marks will meet or exceed Class B Order 2 specifications as set out in the ICSM Standards and Practices for Control Surveys (SP1) . Used for new

survey areas to be proclaimed or for subdivisional development of greater than five (5) lots (subject to negotiation where MGA is not practical).

- iii. Where no suitable control exists in the form of co-ordinated permanent survey marks, Council will provide coordinates of Class B Order 2 within two (2) weeks of receiving a formal request, at a set fee. Refer Appendix N for Request Form.

c. Height Datum

All height information will be on either Australian Height Datum or Australian Height Datum (Derived) and to a minimum standard of Third Order Levelling. Third order levelling is retained because of traditional acceptance. (Refer to Class C Differential Levelling (LC) as defined in the ICSM Standards and Practices for Control Surveys (SP1))

d. Meridian Datum

The meridian datum may be one of the following:

- i. For MGA co-ordinates
 - The meridian is derived from the adjusted survey traverse between the coordinated survey marks.
- ii. For Assumed co-ordinates
 - Azimuth of the current Australian Mapping Grid
 - County Arbitrary Meridian
 - Meridian from an original survey or adjoining survey

2. “AS CONSTRUCTED” DIGITAL GROUND SURVEY

A digital ground survey is required to produce the three-dimensional model of the changes to the natural surface and the location of all artificial features, pursuant to an approved operational works development permit. The Digital Ground Survey will comprise of:-

- a) Sufficient measurements to both topographical features and constructed assets, to ensure that all points and strings in the digital model accurately reflect their true geometric shape and location on the earth's surface.
- b) Spacing between points will ensure that a tolerance for the length of triangle sides in the triangulation model do not exceed 25m on pavement and 50m outside of pavement.

Data provided to Council as part of the “As Constructed” submission shall include the export of the processed survey data in either an ASCII format or in an appropriate format that is requested by the FNQROC member Council that is compatible with their design or survey software.

3. “AS CONSTRUCTED” DRAWINGS

Council requires “As Constructed” Drawings to be produced using “AutoCAD” Software and submitted in DWG format only. Consultants shall ensure that when lodging AutoCAD drawing files that they are compatible with the current version of AutoCAD being used by the relevant member Council. No drawing sheets or title blocks shall be used on the file.

- a) Specific features are represented by blocks and certain linestyles, and such require specific definition and attribute details recorded. These features, their definitions and attribute data requirements are covered in the following sections of this document.
- b) Service plans shall be submitted in Grayscale 'PDF' format, digitally certified or with the certification (**Appendix K**) as per Section CP1.22 subsection 2 and with one (1) additional B&W hard copy paper print, produced for each service on the consultants drawing sheets. Both PDF and hardcopy plans must be legible.

The plans shall be prepared according to the following scales and sheet sizes:-

- i. Stormwater Drainage - 1: 500 (A1 Sheets)
 - ii. Sewerage - 1: 500 (A1 Sheets)
 - iii. Water - 1: 500 (A1 Sheets)
 - iv. Reservoir Sites - 1:200 (A3 Sheets)
 - v. Pump Stations - 1:200 (A3 Sheets)
 - vi. Treatment Plants - 1:200 (A3 Sheets)
- c) Electronic Data shall be supplied on using Councils As Constructed submission portal (<https://www.asconstructed.com/>) using supplied account details, where council does not have a portal CD's, DVD or USB Flash Drive/Memory Stick.
 - d) The "As Constructed" Drawing may be prepared by either the Consulting Engineer or the Surveyor but must comply with the requirements presented herein.
 - e) Survey accuracy to be metres to three decimal points
 - f) The AutoCAD Drawing shall be a single drawing containing seven (7) main elements:
 - i. Cadastral Base – showing property boundaries, easements and Permanent Survey Marks and Survey Control.
 - ii. Topographical Features – including kerbing/edge of seal, top and toe of batters, change of grades, retaining wall, watercourses, structures, landscaping and park embellishments, contours at 0.5m intervals etc.
 - iii. Water – showing: existing infrastructure and the connection details for new infrastructure, offsets from boundary, connection points, main size, valves, hydrant locations to property boundaries etc together with the location of any irrigation pipes and associated fittings, and details of any water infrastructure abandoned or removed as a result of the new works.
 - iv. Sewerage - showing existing infrastructure and the connection details for new infrastructure, pipe invert levels, pipe diameter and grades, cover levels, location to property boundary, distance from downstream manhole to PCB's, PCB's levels and type, and details of any sewerage infrastructure abandoned or removed as a result of the new works.
 - v. Stormwater Drainage - showing pipe invert levels, pipe diameter and grades, pipe material, finished surface levels, drainage structure description, catch drains, open drains/swales etc.
 - vi. Roads –, showing centre of road carriageway, kerbing/edge of seal.
 - vii. Pump and lift stations Specifications as listed in 'Drafting Requirements' – Sewerage Pump Stations

- g) If MGA co-ordinates have been used, “As Constructed” drawings can be submitted in MGA Zone 55 (GDA94) or MGA Zone 55 (GDA2020). The datum used should be clearly noted on drawing details.

4. DRAFTING REQUIREMENTS (“AS CONSTRUCTED”) ^{CRC}

The general drafting requirements for the preparation of “AutoCAD” drawings shall be as detailed in this section. Any elements encountered in the preparation of these drawings not specifically covered by this manual shall be confirmed with Council's Asset Management Section prior to submission of drawing file.

- a) The orientation of the drawing must be set to AutoCAD’s default (ie 90 at 12 o’clock, and anticlockwise measured angles).
 - b) 1 Drawing unit = 1 metre
 - c) All symbols and line types to be as specified within this section. A digital file of Councils linestyles, layers and blocks will be made available.
 - d) All colours are to be by layer (except internal block linework)
 - e) All line types are to be by layer
 - f) All other colours and line widths to reflect FNQROC.ctb file.
 - g) AutoCAD layer names shall be in accordance with Table CP1.2 or as specified within this section.
 - h) All lines are to be 2D poly lines and all blocks are to have a z value (level value).^{MSC}
- (^{MSC}: All lines are to be 3D poly lines and all blocks are to have embedded z value (level value))

Table CP1.2 Layering Standards

Description of Layer	Annotation	AutoCAD Layer	AutoCAD Linetype	AutoCAD Colour Index
2.0 mm high text	2.0 mm	020_TXT	Continuous	254 (light grey)
2.5 mm high text	2.5 mm	025_TXT	Continuous	7 (white)
3.5 mm high text	3.5 mm	035_TXT	Continuous	2 (yellow)
5.0 mm high text	5.0 mm	050_TXT	Continuous	1 (red)
7.0 mm high text	7.0 mm	070_TXT	Continuous	5 (blue)
10.0 mm high text	10.0 mm	100_TXT	Continuous	30 (Orange)
Contours	N/A	CONTOUR	Continuous	252 (dark grey)
Contour heights	2.0 mm	CONTOUR_HEIGHT	Continuous	254 (light grey)
As Constructed Above Ground Electricity	N/A	ELECTRIC_ABOVE	– E – E –	220
As Constructed Underground Electricity	N/A	ELECTRIC_UNDER	– E – – – E –	220
As Constructed Aboveground Telecommunication	N/A	TELECOM_ABOVE	– T – T –	133

Description of Layer	Annotation	AutoCAD Layer	AutoCAD Linetype	AutoCAD Colour Index
As Constructed Underground Telecommunications	N/A	TELECOM_UNDER	– T – – – T –	133
As Constructed Above ground Optical Fibre	N/A	OPTIC_ABOVE	– OF – OF –	133
As Constructed Underground Optical Fibre	N/A	OPTIC_UNDER	– OF – – – OF –	133
As Constructed Fuel Line	N/A	FUEL_LINE	– F – – – F –	44
As Constructed Gas Line	N/A	GAS_LINE	– G – – – G –	23
Electricity text	2.5 mm	ELECTRIC_TXT	Continuous	7 (white)
Telecom Text	2.5 mm	TELECOM_TXT	Continuous	7 (white)
Optical Fibre Text	2.5 mm	OFTIC_TXT	Continuous	7 (white)
Fuel Line Text	2.5 mm	FUEL_LINE_TXT	Continuous	7 (white)
Gas Line Text	2.5 mm	GAS_LINE_TXT	Continuous	7 (white)

5. LINEWORK

a) It should be noted that Pen Size Colours are as follows:

0.15	-	132	
0.25	-	8	(Grey)
0.25	-	4	(Cyan)
0.35	-	7	(White)
0.35	-	200	(Purple)
0.50	-	2	(Yellow)
0.50	-	3	(Green)
0.70	-	5	(Blue)
0.50	-	6	(Magenta)
1.00	-	1	(Red)
1.00	-	30	(Orange)

b) Linetype scale shall be = 10

c) All line types shall be taken from the Regional Organisations standard line type file (FNQROC.LIN) or from the Department of Main Roads – ‘Drafting and Design Presentation Standards’.

6. TEXT STYLES

a) Text styles to be used on all Drawings shall be specified as below:

Text Style Name	Font Name	Height	Width	Oblique Angle	Backwards	Upside Down	Vertical
RS	Romans	0.0	1.0	0d'0"0"	N	N	N
RSO	Romans	0.0	1.0	15d'0"0"	N	N	N

7. BLOCKS

- a) Council supplied blocks shall be used at all times and XREF blocks shall not be used. Many of these blocks will have numerous attributes (visible and hidden) attached to them. It is the responsibility of the Consultant preparing the As Constructed Digital Submission to complete all attributes as identified in the block attributes tables. The remaining attributes are for Council's use. Blocks must be inserted at and must remain at a scale of 1:1.

8. ACCURACY REQUIREMENTS

- a) Dimensions shall be used to accurately define the location of the service entities in the as-constructed data (the dimension requirements are described below). However, to ensure the clarity of the utility plans, Council requires that consultants separate the entities to enable them to be easily identifiable at the appropriate scale. Relativity among the entities and in relation to other features must be maintained (eg if an entity is to the east of a boundary it must be shown on the plan to be east of that boundary).

Location:

- i. Dimensions shall be shown to two decimal place
- ii. Pipe lengths shall be shown to two decimal place.

Level:

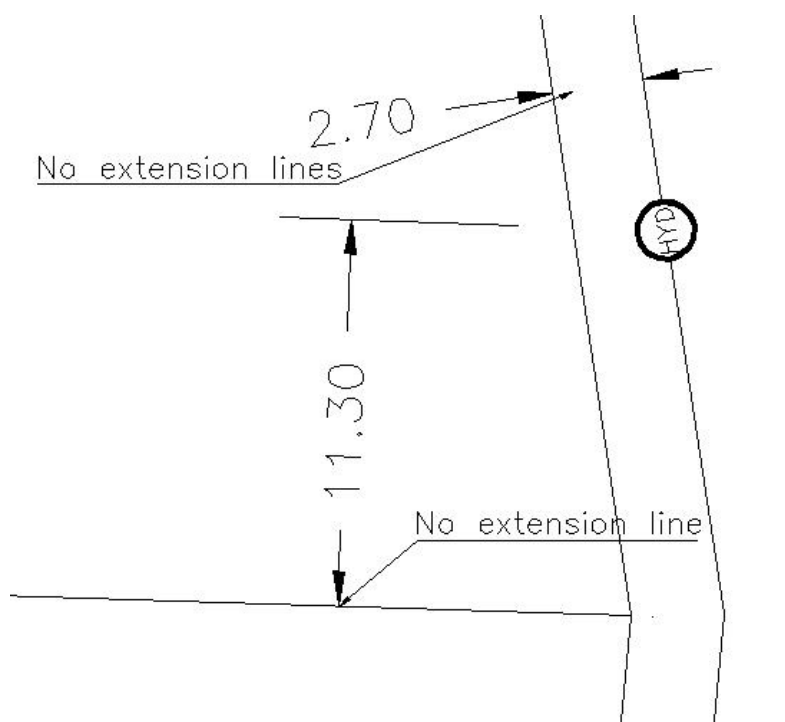
- iii. PCB Invert Levels shall be shown to two decimal places.
- iv. Pipeline Invert Levels shall be shown to two decimal places.
- v. RL Manhole Lid levels shall be shown to two decimal places
- vi. All depths shall be shown to two decimal places.
- vii. Finished Surface Levels shall be shown to two decimal places.
- viii. Pipe Grades shall be expressed as a ratio (eg 1:150) determined from full survey accuracy.

9. DIMENSIONING

- a) Council requires the Dimensioning of “As-Constructed” services to conform to the following criteria:
- i. There must be sufficient dimensions to define the location of the service without ambiguity. Pipelines must be dimensioned sufficiently to show their alignment in relation to the cadastre. Service entities (valves, hydrants, manholes, PCB's, etc) must be located in relation to the nearest cadastral corner. PCB's must be located in relation to the nearest cadastral corner of the lot it services. If PCB's have been installed to service a future stage of development, then PCB details relevant to the future stage lots are to be presented on the as-constructed drawing for that stage . Note: in areas containing a number of service entities clustered together, dimensioning is to be sufficient to locate the main elements of the cluster only (eg every service entity is not required to be fully dimensioned). Service entities located opposite cadastral boundary intersections do not require dimensioning.
 - ii. The dimensioning of the utilities is to be in accordance with the Council's dimensioning styles. These styles are set in the prototype drawing that can be obtained from

Councils Asset Management Section upon request. Council has developed a style for use with each of the utilities (water, sewerage and drainage). The appropriate style to use is listed with the individual utilities' requirements.

- iii. Dimension extension lines must be created manually in the appropriate dimensions layer. Extension lines are not to be created that cover other linework (pipelines, property boundaries etc). See diagram below. Note: the dimension styles supplied by Council have the extension lines set to none as a default. Do not reset
- iv. Dimension text is to be outside of the extension lines and clear of the roadways. See diagram below.
- v. Dimensions are to be layered separately for each utility. Please refer to the relevant utilities requirements for the appropriate layer.



10. SURVEY CONTROL

The Survey Control entity requirements are as follows:

a) Linework

Description	Layer	Colour (by layer)	Linetype (by layer)
Survey Traverse	AC_SURVEY_TRAVERSE	Dark Green (96)	Continuous

b) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Permanent Survey Mark	AC_SURVEY_PSM	Dark Red (12)	RS	1.25
Survey Instrument Station	AC_SURVEY_IS	Dark Red (12)	RS	1.25

Survey Traverse details	AC_SURVEY_TRAVERSE	Dark Green (96)	RS	1.25
-------------------------	--------------------	-----------------	----	------

c) Blocks

Block	Name	Description	Layer	Colour
	SURVEY_PSM	Permanent Survey Mark	AC_SURVEY_PSM	12 (Dark Red)
	SURVEY_INST	Instrument Station	AC_SURVEY_INST	12 (Dark Red)

- d) Bearings and Distances shall be recorded against all sections of the traverse line in the AC_SURVEY_TRAVERSE layer.

11. CADASTRAL BASE

The Cadastral Base entity requirements are as follows:

a) Linework

Description	Layer	Colour (by layer)	Linetype (by layer)
Property Boundary	AC_CADASTRE_PROPBDY	4 (Cyan)	Continuous
Easements	AC_CADASTRE_EASE	4 (Cyan)	Dashed (0.0 wide)
Existing Boundary	AC_CADASTRE_EXBDY	8 (Grey)	Continuous
Existing Easements	AC_CADASTRE_EXEASE	8 (Grey)	Dashed (0.0 wide)
Future Boundary	AC_CADASTRE_FUTBDY	8 (Grey)	Continuous
Future Easement	AC_CADASTRE_FUTEASE	8 (Grey)	Dashed (0.0 wide)
Major Contours	AC_MAJOR_CONTOURS	252 (Dark Grey)	Contour Major
Minor Contours	AC_MINOR_CONTOURS	254 (Light Grey)	Contour Minor
Note: Contour information will be for internal council use only. Not required on plotted hard copies.			

- i. Property boundary linework shall not be broken when crossed by text. All text is to be located clear of linework whenever possible. Refer to the Table above.
- ii. All boundaries between allotments and road reserve will be placed in the AC_CADASTRE_PROPBDY layer.
- iii. Each parcel to be an individual closed polyline

b) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Permanent Survey Mark	AC_CADASTRE_PSM	4 (Cyan)	RS	1.25
Lot No's	AC_CADASTRE_LOTNO	2 (Yellow)	RS	1.25
Registered Plan No's	AC_CADASTRE_RPNO	3 (Green)	RSO	2.0
Drainage Reserve Text	AC_CADASTRE_DRTEXT	4 (Cyan)	RS	1.25
Contour Text	AC_CONTOURS	252 (Dark Grey)	RS	1.0

- i. Allotment numbers are to be located in the centre of the boundary opposite the sewer line and inside the allotment (eg if the sewer line is along the rear boundary, numbers should be placed centrally along the front boundary) or close by in a clear position.
- ii. All text shown on plan, other than acceptable abbreviations, shall be in uppercase.

12. WATER RETICULATION ^{CRC}

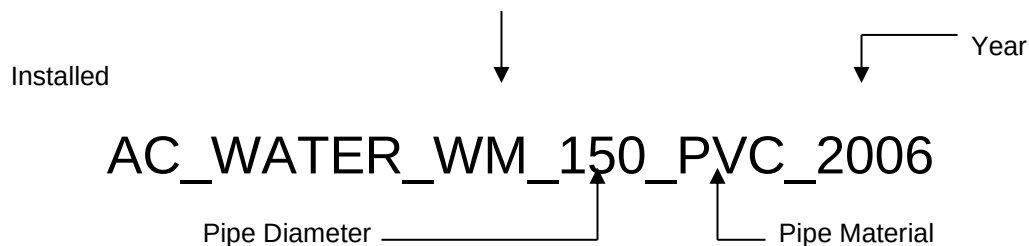
The Water entity requirements are as follows:

a) Linework ^{CRC}

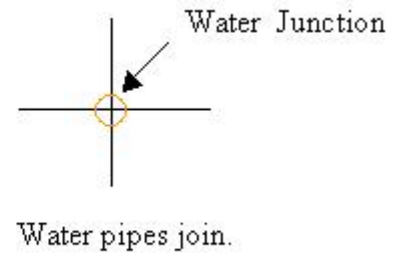
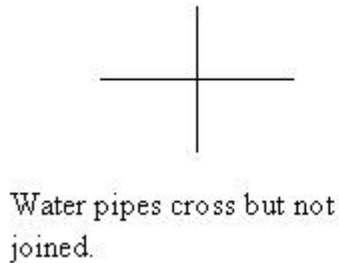
Description	Layer	Colour (by layer)	Linetype (by layer)
Water Main > 50 dia	Refer Below (Except CRC, use AC_WATER_PIPE)	2 (Yellow)	Continuous Polyline (0.0 wide)
Water Main ≤ 50 dia	All others Refer Below (Except CRC, use AC_WATER_PIPE)	4 (Cyan)	Continuous Polyline (0.0 wide)
Trunk Main	Refer Below (Except CRC, use AC_WATER_PIPE)	2 (Yellow)	Dashed Polyline (0.0 wide)
Private Service	Refer Below (Except CRC, use AC_WATER_PIPE)	4 (Cyan)	Continuous Polyline (0.0 wide)
Reclaimed Water Main	Refer Below (Except CRC, use AC_WATER_PIPE)	200 (Purple)	Continuous Polyline (0.0 wide)
Service to Park or Landscaping	Refer Below (Except CRC, use AC_WATER_PIPE)	7 (White)	Continuous Polyline (0.0 wide)
Existing Water Features	AC_WATER_EXIST	8 (Grey)	Dashed (0.0 wide)

b) Layer Names for water mains shall be in the format shown below (except for CRC):

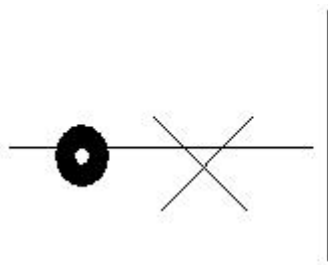
Denotes Water Reticulation Type: WM – Water Main, WT – Trunk Main, WP – Private Main, WI – Irrigation and WR – Reclaimed Water



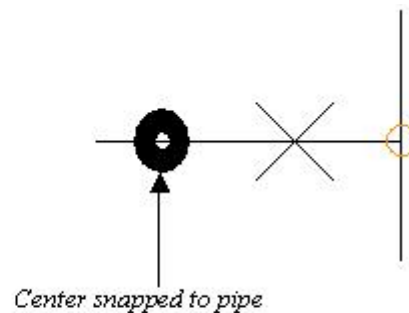
- c) Crossing and connecting water pipes are to be shown as below. Connecting water pipes are to be represented by the Water_Junction block.



- d) Water mains and irrigation pipes are to be one continuous 2D ^(MSC: 3D) Polyline, broken only at pipe junctions and valves.
- e) Ensure that all pipes/ junctions/ valves are snapped to the centre/ end of the object.



Incorrect



Correct

- f) Connections to existing water features in Council controlled land are required and are to be placed in the AC_WATER_EXIST layer.
- g) Text ^{CRC}

Description	Layer	Colour	Text Style	Text Height
Pipe Size, Material and Notes	AC_WATER_TEXT (Except CRC, use AC_WATER_PIPE)	4 (Cyan)	RS	2.0
Reservoir Text	AC_WATER_TEXT	4 (Cyan)	RS	2

- h) Dimensioning
- Dimension Style - WATER
 - Dimension Layer – AC_WATER_DIM

- i) The following details must be shown on the 'as constructed' drawing :
- a. Mains
 - i. End caps, bends, fittings and thrust blocks
 - ii. Location – Above or below ground, default below
 - iii. Length – Length of pipe measured between junctions or changes in horizontal direction. (if requested)
 - iv. Installation depth, if not standard.
 - v. Dimensions from horizontal bends in the main to two (2) property boundaries or corners
 - vi. Operating pressure at critical locations (low points, high points). (If requested)
 - vii. Location of valves, air vents, scour valves, bends, access chambers and hydrants along the main
 - viii. For trunk mains: Long sections with surface level and crown level at changes of grade, and at a maximum of 30 metre centres.
 - b. Reservoirs
 - i. Site locality plan showing the reservoir and associated works in relation to cadastral boundaries
 - ii. Significant variations from the approved drawings including tolerances outside those specified
 - iii. Actual dimensions of reservoir
 - iv. Actual configuration of associated pipework, ladders, walkways, control boxes, man access hatch etc
 - v. Location of water service
 - vi. Set out, configuration and details of scour and overflow pipework
 - vii. Location and details of any driveways, apron slabs, fences and any other utilities.
 - c. Pump Station
 - i. Actual dimensions of building
 - ii. Control Pressure - Pump control pressure.
 - iii. Control Capacity – Pump control capacity
 - iv. System curves and pump curves.
 - v. Number – Number of pumps Installed. The pumps are individually represented by the Pump block and placed next to the pump station.
 - vi. Valve Pit Floor Level, lid level and outside FSL in metres, A.H.D
 - vii. Cross reference detail drawings and standard drawings
 - viii. Full pump specification include duty, make, model, motor rating, curve number, impeller, diameter, etc.
 - ix. Site locality plan at (scale 1:200) showing the as-constructed pump station and associated works in relation to cadastral boundaries
 - x. Location of water service

- xi. Location of operation and maintenance manuals
- xii. Finished surface contour levels and spot levels
- xiii. Clearance dimensions between flanges and wall surfaces
- xiv. Cover to pipe work
- xv. Switch Board Layout Details
- xvi. Actual internal configuration of pumps, pipework, control panel, generator etc
- xvii. Q100 flood line level and highest Recorded Flood Level in metres, A.H.D.
- xviii. Locations and details of any driveways, apron slabs, fences and any other utilities.

d. Pumps

- i. Installation Date - Year in which the asset was constructed or installed i.e. 2007
- ii. Number – Pump serial number.
- iii. Housing – Housing material.
- iv. Impeller – Impeller material
- v. Impeller Diameter
- vi. Impeller curve
- vii. Shaft size
- viii. Shaft seal – Type of seal used.
- ix. Flow Rate.
- x. Motor Kilowatts
- xi. Motor Current
- xii. Motor control (VSD or DOL)
- xiii. Bearing Size.
- xiv. Size – Pump Size (weight and outside dimensions).

e. Electrical Controls

Required Attributes:

- i. Iso_size – Main Isolator size.
- ii. Type – Isolator Type.
- iii. Manufacturer – Isolator manufacturer.
- iv. Inc_Size – Main Incomer size.
- v. Wiring_enclosure – whether wiring enclosure is buried direct, conduit, cable tray.
- vi. Voltage – Voltage of switchboard.
- vii. IP – IP rating.
- viii. Installation Date - Year in which the asset was constructed or installed i.e. 2007.

- ix. Reference Dwg – Drawing number for design of constructed asset.
- x. Reference Manual – Manual detailing components etc of switchboard.

f. Valve Pit / Chlorination Plant

Required Attributes:

- i. Installation Date - Year in which the water asset was constructed or installed i.e. 2007
- ii. Reference Dwg – Drawing number for design of constructed water asset

g. Water Block Details

Block	Name	Description	Layer	Colour
	Water_Air	Air Valve	AC_WATER_AV	2 (Yellow)
	Water_Chlorin	Chlorination Plant	AC_WATER_WC	4 (Cyan)
	Water_Elec	Electrical Controls	AC_WATER_EC	4 (Cyan)
	Water_End	Endcap for pipes	AC_WATER_END	2 (Yellow)
	Water_Float	Float Valve	AC_WATER_FV	2 (Yellow)
	Water_Hydraulic	Hydraulic / Control Valve	AC_WATER_HV	2 (Yellow)
	Water_Hydrant	Fire Hydrant	AC_WATER_HYD	2 (Yellow)
	Water_Junction	Junction of Water Pipes	AC_WATER_JUNC	30 (Orange)
	Water_Pressure	Pressure reducing Valve	AC_WATER_PRV	2 (Yellow)
	Water_Reducer	Reducer	AC_WATER_RED	2 (Yellow)
	Water_Reflux	Reflux Valve	AC_WATER_RFV	2 (Yellow)
	Water_Res	Reservoir	AC_WATER_RES	2 (Yellow)
	Water_Scour	Scour Valve	AC_WATER_SCRV	2 (Yellow)
	Water_Stop	Stop Valve	AC_WATER_SV	2 (Yellow)
	Water_VPit	Valve Pit	AC_WATER_VP	4 (Cyan)
	Water_Meter	Water Meter	AC_WATER_WM	4 (Cyan)
	Water_PStn	Water Pump Station	AC_WATER_PS	4 (Cyan)

13. SEWERAGE RETICULATION ^{CRC}

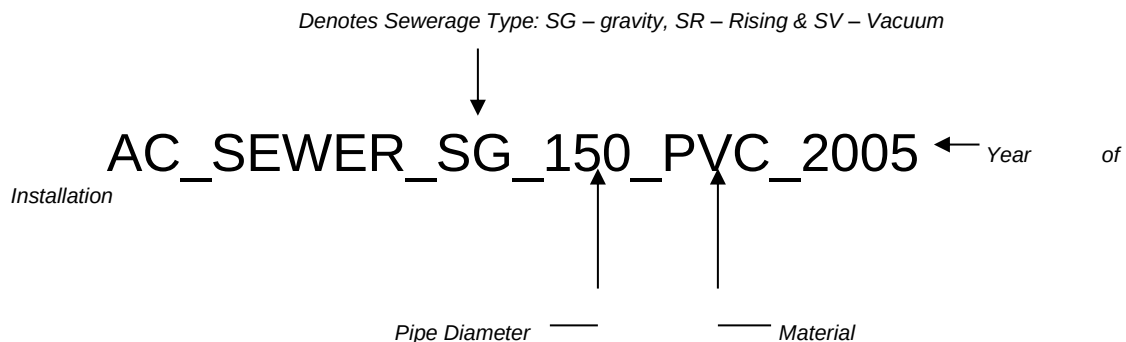
The Sewer entity requirements are as follows:

a) Linework ^{CRC}

Description	Layer	Colour (by layer)	Linetype (by layer)
Pressure Rising(Pressure) (Rising) Mains	Refer Below (Except for CRC, use AC_SEWER_RPIPE)	6 (Magenta)	Pmain (assign linetype to object)
Gravity Mains	Refer Below (Except for CRC, use AC_SEWER_RPIPE)	6 (Magenta)	Continuous
Vacuum Mains	Refer Below (Except for CRC, use AC_SEWER_RPIPE)	6 (Magenta)	Vmain (assign linetype to object)
Existing Features	AC_SEWER_EXIST	8	Dashed (0.0 wide)
*PCB's	AC_SEWER_PCB	Red	Continuous
Property Sewers (Private)	AC_SEWER_PROPERTY	Red	Continuous

* Refers to all PCB's, which are to service new lots created by the development to which the as-constructed drawing relates, regardless of whether or not the PCB's were installed in a previous construction or installed on an existing sewer. (Where existing PCB's have been obtained from a third party, it is recognised that they cannot be certified and in such cases need to be noted.)

- b) Connections to existing sewer features in Council controlled land are required and are to be placed in the AC_SEWER_EXIST layer.
- c) Sewer pipes are to be one continuous 2D^(MSC: 3D) polyline between manholes/valves/pump or lift stations.
- d) Layer Names for Mains shall be in the format shown below (except for CRC):

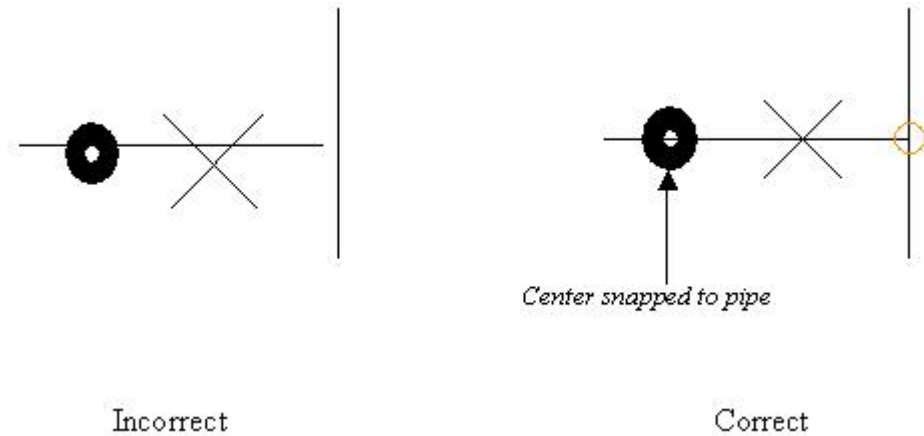


e) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Pipe Dia, Material, Length, Grade & Invert Levels	AC_SEWER_PIPE	Cyan	RS	1.25
PCB Details	AC_SEWER_PCB	Cyan	RS0	1.0

- All text other than acceptable abbreviations shall be in uppercase.

- f) Ensure that all pipes/ junctions/ valves are snapped to the centre/ end of the object.



- g) Dimensioning

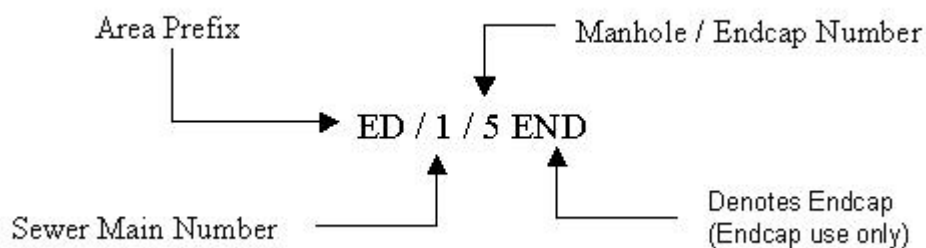
- Dimension Style - SEWER
- Dimension Layer - AC_SEWER_DIM

- h) Sewer Block Details

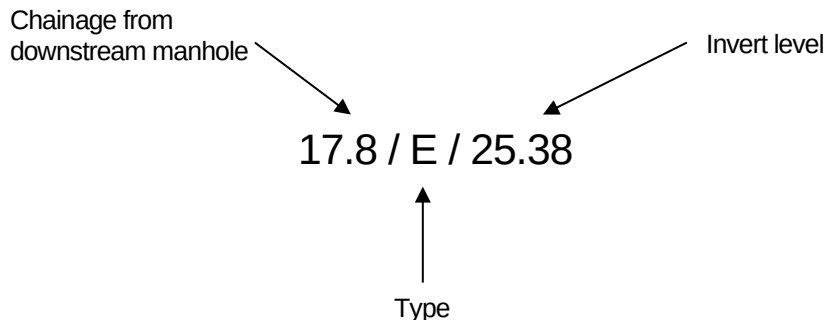
Block	Name	Description	Layer	Colour
	Sewer_Air	Air Valve	AC_SEWER_AV	3 (Green)
	Sewer_End	Endcap for pipes	AC_SEWER_END	Magenta
	Sewer_FSL	Finished Surface Level	AC_SEWER_FSL	4 (Cyan)
 CHAMAGE/TYRE/L	Sewer_PCB	Property Connection	AC_SEWER_PCB	6 (Magenta)
	Sewer_Reflux	Reflux Valve	AC_SEWER_RFV	3 (Green)
	Sewer_Scour	Scour Valve	AC_SEWER_SCRV	3 (Green)
	Sewer_IO	Inspection Openings	AC_SEWER_IO	4 (Cyan)
	Sewer_MS	Sewer Maintenance Shaft	AC_SEWER_MH	4 (Cyan)
	Sewer_MH	Sewer Manhole	AC_SEWER_MH	4 (Cyan)
	Sewer_ScMH	Sewer Scour Manhole	AC_SEWER_MH	4 (Cyan)

Block	Name	Description	Layer	Colour
	Sewer_PStn	Sewer Pump Station	AC_SEWER_PS	Magenta
	Sewer_Stop	Sewer Stop Valve	AC_SEWER_SV	3 (Green)
	Sewer_Elec	Electrical Controls	AC_SEWER_EC	4 (Cyan)
	Sewer_VPit	Valve Pit	AC_SEWER_VP	4 (Cyan)
	Sewer_VacStn	Vacuum Pump Station	AC_SEWER_VACPS	3 (Green)
	Sewer_Store	Storage Tank	AC_SEWER_ST	4 (Cyan)
	Sewer_Oflow	Overflow Pit	AC_SEWER_OP	4 (Cyan)
	Sewer_VacPit	Vacuum Pit	AC_SEWER_VACPT	4 (Cyan)
	Sewer_Reducer	Reducer	AC_SEWER_RED	3 (Green)
	Sewer_Pipe	Pipe attribute detail	AC_SEWER_PIPE	4 (Cyan)

- i) The preferred method of denoting pipe invert levels for all sewer pipes is via the use of Sewer_Pipe Block.
- j) Manhole number shall be allocated as shown below:



- k) Endcaps and rodding points use a very similar format. Endcaps must have the word "END" included after the endcap number (eg ED/26/6 END) and rodding points must have the letters "RP" included as part of the rodding number (eg ED/26/RP6).
- l) Area prefix numbers are listed in the Local Government Specific section
- m) Property Connection Branches shall be described using the following format. Refer to Standard drawing S3005 for different PCB types.



Note: For Type A the invert level will be taken as the invert level of the IO pipe of the branch (refer S3005). All other types are invert level of the sewer main at the PCB.

n) Finished Surface Levels

- a. Finished surface levels are required at all cadastral corners. The Sewer_FSL block must be used for all finished surface level information. The surface level data at cadastral corners must be placed in the layer AC_SEWER_FSL .

o) Pipe Invert Levels

- a. Gravity Mains - Invert levels must be provided for all gravity mains. The pipe invert levels are to be recorded in the Sewer Pipe Block.
- b. Pressure (Rising) Mains - Invert level information is required along all pressure (rising) mains. They must be shown at valves, in the pump station and at discharge points. The pipe invert levels are to be recorded in the Sewer Pipe Block.

p) The following details must be shown on the 'as constructed' drawing:

- a. Pressure Sewer Rising Main
 - i. Operating pressures at low and high points
 - ii. Line velocity at maximum operating flow
 - iii. Surface levels and invert levels at ends, changes in grade and at a maximum of 30 metre centres on a long section.
 - iv. Running chainages from the pump station to changes of grade, valves, air vents, scour valves, bends and access chambers along the main
 - v. Dimensions from horizontal bends in the main to two (2) property boundaries or corners

- q) For pump and lift stations, all relevant details are to be recorded in the relevant pump station block. Refer to block for required data.

14. SEWERAGE PUMP STATIONS

- a) The pump/lift station number shall be provided by Council.
- b) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Pipe Dia, Material, Length, Grade & Invert Levels	AC_SEWER_PS	4 (Cyan)	RS	1.25
Misc. text descriptions	AC_SEWER_TEXT	4 (Cyan)	RSO	1.0

- c) The following details must be shown on the 'as constructed' drawing:

- a. Storage/Overflow
- i. Site locality plan showing the as-constructed storage infrastructure and associated works in relation to cadastral boundaries
 - ii. Show all significant variations from the approved drawings including tolerances outside those specified
 - iii. Actual dimensions of storage infrastructure
 - iv. Actual set out and configuration of piping and external to the storage infrastructure.
 - v. Q100 flood line level and Highest Recorded Flood level in metres, A.H.D
 - vi. Overflow discharge location
 - vii. Location of water services
 - viii. Finished surface contour levels and spot levels
- b. Pump and Lift Stations
- i. Locations and details of any driveways, apron slabs, fences and utilities.
 - ii. Q100 flood line level and Highest Recorded Flood Level in metres, A.H.D.
 - iii. Full pump specification including duty, make, model, motor rating, curve number, impeller, diameter, etc. (shown as a table on the plan)
 - iv. Site locality plan at (Scale 1:200) showing the as-constructed pump station, valve box and associated works in relation to cadastral boundaries

- v. Overflow discharge location
 - vi. Location of water service
 - vii. Location of operation and maintenance manuals
 - viii. Finished Surface Contour Levels and spot levels
 - ix. Clearance dimensions between flanges and internal wall surfaces
 - x. Cover to pipe work
 - xi. Show existing and proposed Manholes within the area of the locality plan including manhole number
 - xii. Switch Board Layout Details and any telemetry (if applicable)
 - xiii. All significant variations from the approved drawings including tolerances outside those specified.
 - xiv. System Curve – System head pressure.
 - xv. Pump Curve – Pump design head
 - xvi. Head – Maximum head at receiving mains.
 - xvii. Underside of plug (ballast) in metres, A.H.D
 - xviii. RL at Top of Sewer Pump Station
 - xix. Invert Level of Pressure Main
 - xx. Invert Level of each Gravity Main
 - xxi. Invert Level of Overflow Pipe
 - xxii. RL of High Level Alarm
 - xxiii. RL of Standby Start
 - xxiv. RL of Pump Duty Start
 - xxv. RL of Pump Stop
 - xxvi. RL of Low Level Alarm
 - xxvii. RL of Pump Station floor
 - xxviii. Pump Station Internal Diameter
- c. Electrical Controls and Pumps

Required Attributes:

- i. Refer to attribute details for sewer electrical controls and pumps.

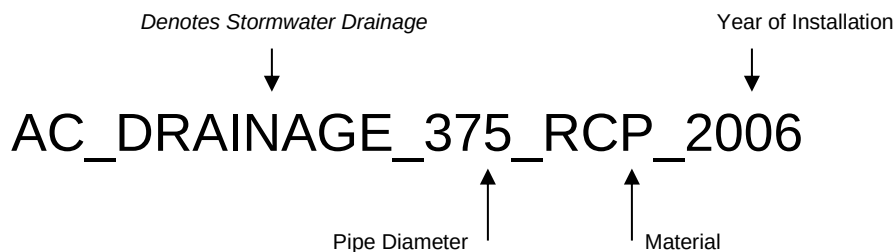
15. STORMWATER DRAINAGE SYSTEM^{CRC}

The Stormwater entity requirements are as follows:

a) Linework^{CRC}

Description	Layer	Colour (by layer)	Linetype (by layer)
RC Pipes	Refer below (Except for CRC, use AC_DRAINAGE_PIPE)	1 (Red)	RCP (Polyline 0.5 wide)
PVC Pipes	Refer below (Except for CRC, use AC_DRAINAGE_PIPE)	1 (Red)	Continuous (Polyline 0.25 wide)
Box Culverts	Refer below (Except for CRC, use AC_DRAINAGE_PIPE)	7 (White)	RCBC (Polyline 0.5 wide)
Catch Drain	Refer below (Except for CRC, use AC_DRAINAGE_PIPE)	1 (Red)	CDP (Polyline 0.5 wide)
Sub Surface Drains	AC_DRAINAGE_SSURF	3 (Green)	SSD (Polyline 0.0 wide)
Retaining Walls	AC_DRAINAGE_RETWALL	132	Continuous (Polyline 1.0 wide)
Existing Features	AC_DRAINAGE_EXIST	8 (Grey)	Dashed (0.0 wide)
House drainage (private)	AC_SEWER_HOUSE	8 (Grey)	Continuous

- b) Connections to existing stormwater features in Council controlled land are required and are to be placed in the AC_DRAINAGE_EXIST layer.
- c) Layer names for stormwater lines shall be in the format shown below (except for CRC):



- d) Catch Drain polylines are to be a 2D^(MSC:3D) polyline and based on the centre of the constructed drain.

e) Text^{CRC}

Description	Layer	Colour (by layer)	Text Style	Text Height
Catch Drain Levels	AC_DRAINAGE_TEXT	4 (Cyan)	RS	1.0
RCP Diameter, Material, Length and Invert Levels	AC_DRAINAGE_TEXT	4 (Cyan)	RS	1.0
Box Culvert, Size, Material Length & Invert Levels	AC_DRAINAGE_TEXT	4 (Cyan)	RS	1.0
Sub Surface Drain Text	AC_DRAINAGE_TEXT	4 (Cyan)	RS	1.0
Kerb Inlet Type & RL Top	AC_DRAINAGE_KERB	4 (Cyan)	RS	1.0
Manhole & RL Top	AC_DRAINAGE_MH	4 (Cyan)	RS	1.0
Endwalls	AC_DRAINAGE_HDWALL	4 (Cyan)	RS	1.0
Field Inlet Pit	AC_DRAINAGE_PIT	4 (Cyan)	RS	1.0
Finished Surface Level	AC_DRAINAGE_FSL	132	RS	1.0

f) Dimensioning

- a. Dimension Style - DRAINAGE
- b. Dimension Layer – AC_DRAINAGE_DIM

g) Finished Surface Levels

- h) Finished surface levels are required at all cadastral corners and at changes of grade. The AC_DRAINAGE_FSL block must be used for all finished surface level information and placed in the layer AC_DRAINAGE_FSL.

- i) Both above ground and underground drainage details are to be recorded in the AC_DRAINAGE_TEXT layer.

j) Drainage Blocks

Block	Name	Description	Layer	Colour
	Drain_Culvert	Culvert Text Box	AC_DRAINAGE_CULVERT	4 (Cyan)
	Drain_FSL	Finished Surface Level	AC_DRAINAGE_FSL	4 (Cyan)
	Drain_GPT	Gross Pollutant Trap	AC_DRAINAGE_GPT	4 (Cyan)
	Drain_IO	Inspection Opening	AC_DRAINAGE_IO	4 (Cyan)
	Drain_HW	Headwall	AC_DRAINAGE_HDWALL	4 (Cyan)
	Drain_EWW	Head and Wingwalls	AC_DRAINAGE_HDWALL	4 (Cyan)
	Drain_RKerb	Kerb Inlet Pit/ Grate located RHS	AC_DRAINAGE_KERB	4 (Cyan)
	Drain_MKerb	Kerb Inlet Pit/Grate located Middle	AC_DRAINAGE_KERB	4 (Cyan)
	Drain_LKerb	Kerb Inlet Pit/Grate located LHS	AC_DRAINAGE_KERB	4 (Cyan)
	Drain_MH	Manhole	AC_DRAINAGE_MH	4 (Cyan)
	Drain_Open	Open Drain Text Box	AC_DRAINAGE_DRAIN	4 (Cyan)
	Drain_Pipe	Pipe Text Box	AC_DRAINAGE_PIPE	4 (Cyan)
	Drain_Pit	Field Inlet Pit	AC_DRAINAGE_PIT	4 (Cyan)
	Drain_CPit	Field Inlet Pit, Concrete Shute	AC_DRAINAGE_PIT	4 (Cyan)
	Drain_PStn	Drain Pump Station	AC_DRAINAGE_PS	4 (Cyan)
	Drain_Tide	Tide Flap/Gate	AC_DRAINAGE_TIDE	4 (Cyan)

k) Location of Structure Point and Levels

Drainage structure location point for the various structures shall be as shown.



GRADED KERB INLET PIT



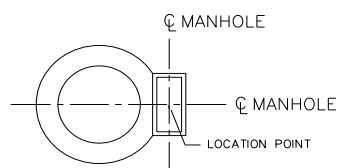
FIELD INLET PITS



MANHOLES



HEADWALLS



GROSS POLLUTANT TRAPS

l) The following details must be shown on the 'as constructed' drawings:

a. Electrical Controls / Pumps

i. Refer to details for water electrical controls and pumps.

16. ROAD NETWORK

The Road Network entity requirements are as follows:

a) Linework

Description	Layer	Colour (by layer)	Linetype (by layer)
Barrier Kerb & Edge Restraint	AC_ROAD_KERB_SKNC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Barrier Kerb & Channel	AC_ROAD_KERB_SKCC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Maintenance Strip	AC_ROAD_KERB_MEDNC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Semi Mountable & Layback	AC_ROAD_KERB_SMKNC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Layback & Channel	AC_ROAD_KERB_SMKCC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Concrete Invert	AC_ROAD_KERB_NKCC	4 (Cyan)	Continuous (Polyline 0.0 wide)
Bridge Deck	AC_ROAD_BRIDGE	6 (Magenta)	Continuous (Polyline 0.0 wide)
Paths	AC_ROAD_PATH	4 (Cyan)	Continuous (Polyline 0.0 wide)
Existing Features	AC_ROAD_EXIST	8 (Grey)	Dashed (0.0 wide)
Thresholds	AC_ROAD_THOLD	4 (Cyan)	Continuous (Polyline 0.0 wide)
Centreline	AC_ROAD_CLINE	132	Continuous (Polyline 0.0 wide)

b) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Kerb & Channel	AC_ROAD_KERB	4 (Cyan)	RS	1.0
Path Material	AC_ROAD_PATH_TEXT	1 (Red)	RS	2.0
Threshold Material	AC_ROAD_THOLD	1 (Red)	RS	2.0
Pavement Details	AC_ROAD_PAV	1 (Red)	RS	2.0
Road Names	AC_ROAD_TEXT	1 (Red)	RS	4.0

c) All constructed footpaths are to be located and recorded in the AC_ROAD_PATH layer, with path material type recorded in the AC_ROAD_PATH_TEXT layer.

d) All different kerb types are to be individually identified by their respective Layer Names refer to standard kerb drawings S1000 & S1001.

- e) All Thresholds are to be represented by a closed polyline, with threshold material type recorded in the AC_ROAD_THOLD layer.
- f) All bridge decks are to be located and recorded in the AC_ROAD_BRIDGE layer.
- g) Subgrade CBR, Road pavement and surface details are to be recorded in the AC_ROAD_PAV layer in the Pavement block
- h) Road Block Details

Block	Name	Description	Layer	Colour
	Traffic_Sign	MUTCD Traffic Signs	AC_ROAD_SIGN	6 (Magenta)
BRIDGE	Bridge	Bridge details	AC_BRIDGE	6 (Magenta)
SUR_MATRL SUR_DEP_CBR BASE_MATRL BASE_DEP_CBR SBASE_MATRL SBASE_DEP_CBR	Pavement	Pavement details	AC_ROAD_PAV	

- i) The following details must be shown on the 'As constructed' drawings:

a. Bridge

- i. Type – Type of bridge, i.e. Road or Pedestrian.
- ii. Deck – Deck material.
- iii. Span – Number of bridge spans.
- iv. Width – Width of bridge decking.
- v. Length – Length of Bridge decking.
- vi. Pylon – Pylon material.
- vii. Headstock – Headstock material.
- viii. WC – Wearing course type, for road bridges.
- ix. Depth – Wearing Course depth, for road bridges.
- x. Installation Date - Year in which the bridge was constructed or installed i.e. 2007.

17. PARKS / LANDSCAPING / STRUCTURES

The Park / Landscaping / Structure entity requirements are as follows:

a) Linework

Description	Layer	Colour (by layer)	Linetype (by layer)
Vegetation	AC_LAND_VEGETATION	96 (Dark Green)	Continuous (Polyline 0.0 wide)
Soft fall Area	AC_LAND_SOFTFALL	30 (Orange)	Continuous (Polyline 0.0 wide)
Paving / Concrete	AC_LAND_PAVING	4 (Cyan)	Continuous (Polyline 0.0 wide)
Building / Structure	AC_LAND_STRUCTURE	4 (Cyan)	Continuous (Polyline 0.0 wide)
Pedestrian Bridge	AC_LAND_BRIDGE	2 (Yellow)	Continuous (Polyline 0.0 wide)

b) Text

Description	Layer	Colour (by layer)	Text Style	Text Height
Vegetation	AC_LAND_TEXT	96 (Dark Green)	RS	1.0
Soft fall Area	AC_LAND_TEXT	96 (Dark Green)	RS	1.0
Lighting Electrical	AC_LAND_TEXT	96 (Dark Green)	RS	1.0
Paving Details	AC_LAND_TEXT	96 (Dark Green)	RS	1.0

c) All distinct landscaping features within the development are to be represented by a closed polyline and on their respective layers. These features shall include but not limited to:

- a. Garden Beds – AC_LAND_VEGETATION, including areas of vegetation with medians and roundabouts.
- b. Areas of pavers / concrete that are not part of a pathway – AC_LAND_PAVING and attribute details recorded in the attribute block Structure.
- c. Soft fall areas – AC_LAND_SOFTFALL and the description of both the border edge and soft fall material in the attribute block Park_Misc.

d) All individual features within the development are to be represented by their respective blocks. These shall include but not limited to:

- Individual Trees.
- Rubbish bins.
- Water bubblers.
- Play Ground Equipment.
- Park Embellishments.

e) The outer edge of all constructed structures are to be located and represented by a closed polyline. In association with the polyline the relevant block will be used to record the attribute details for that structure and placed at the centre of the structure. These structures shall include but not limited to:

- Buildings / Shelters – Attribute block, Structure.
- Pedestrian Bridge – Attribute block, Bridge, refer road section.
- Playing Court, such as basketball or tennis – Attribute block, Park_Misc.

f) Parks / Landscaping Block Details

Block	Name	Description	Layer	Colour
	TREE	Single Tree	AC_LAND_VEGETATION	92
	WATER_FNTN	Water Drinking Fountain	AC_LAND_STRUCTURE	4 (Cyan)
	RUBBISH	Rubbish Bin	AC_LAND_STRUCTURE	4 (Cyan)
	PLAY_EQUIP	Play Equipment	AC_LAND_STRUCTURE	4 (Cyan)
	PARK_MISC	Landscape Embellishments	AC_LAND_STRUCTURE	4 (Cyan)
	STRUCTURE	Building / Shelter	AC_LAND_STRUCTURE	4 (Cyan)

18. UTILITIES

Any utility laid outside of the service trench, i.e. gas or electricity that connects to a Council asset in a park or reserve (other than road reserve), must be located and recorded on layer AC_LAND_UTILITY with reference to linestyles, text and symbols as per AS1100.401 – 1984 “Engineering survey and engineering survey design drawing”

19. ATTRIBUTE INFORMATION REQUIREMENTS (“AS CONSTRUCTED”)
a) WATER

Attribute Information is to be supplied for all new water reticulation assets which ultimately become the property and responsibility of Council. Attribute information is recorded in the water layer format and the various blocks as listed in the “As Constructed” Drawing Requirements water section.

a. Mains

Compulsory attributes:

- i. Nominal Pipe diameter
- ii. Pipe material.
- iii. Pipe Class

Desirable Attributes:

- iv. Pipe lining material (If applicable)
- v. Pipe protection (if applicable)
- vi. JTYPE – Joint type
- vii. Year of Installation.

b. Valves and Hydrants

A valve entity is classified as any sort of flow controlling or limiting device that is attached to a water reticulation or service line. Entities will include but not limited to air valves, sluice valves, scour valves, reducers, end caps and hydrants.

Compulsory Attributes:

- i. Size – Internal Diameter of pipeline valve/ hydrant is connected too. Reducer size shown as 250/150.
- ii. Open/Closed – Whether valve is open or closed

Desirable Attributes:

- iii. Type – Type of flow control entity i.e. stop valve, scour valve, air valve
- iv. Pressure Setting - Pressure Setting in kiloPascal's
- v. Valve off Direction – Direction to turn off valve i.e. Left/Right
- vi. Service – Whether the valve is on a potable or non-potable supply
- vii. Installation Date – Year in which the water asset was constructed or installed i.e. 2007

c. Reservoir

Compulsory Attributes:

- i. Level of floor in metres, A.H.D.
- ii. Capacity – Maximum storage capacity in mega litres

Desirable Attributes:

- iii. Real Property description
- iv. FSL – Finished surface level of the reservoir pad in metres A.H.D
- v. BWL – Bottom water level in metres A.H.D
- vi. TWL – Top water level in metres A.H.D
- vii. Diameter - Internal
- viii. Installation Date – Year in which the water asset was constructed or installed i.e. 2007

d. Water Meter

Compulsory Attributes:

- i. Type – Whether meter is District or Service.
- ii. Meter Number – Unique number assigned to meters by Council.
- iii. Size – Internal Diameter of pipeline meter is connected too.

Desirable Attributes:

- iv. Bore Size – Internal diameter of meter bore size.
- v. Installation Date - Year in which the water asset was constructed or installed i.e. 2007.

e. Pump Station

Compulsory Attributes:

- i. Council Identification Number
- ii. RTU number (if applicable)

Desirable Attributes:

- iii. Inlet – pipe inlet diameter.
- iv. Outlet – pipe outlet diameter.
- v. Head – System head pressure.
- vi. Flow – Design flow rate.
- vii. Installation Date - Year in which the water asset was constructed or installed i.e. 2007
- viii. Slab Level in metres, A.H.D
- ix. Real Property description
- x. Reference Dwg – Drawing number for design of constructed water asset

b) SEWER

Attribute Information is to be supplied for all new waste water assets which ultimately become the property and responsibility of Council. Attribute information is recorded in the sewer layer format and the various blocks as listed in the “As Constructed” Drawing Requirements sewer section.

a. Valves

A valve entity, is classified as any sort of flow controlling or limiting device that is attached to a sewer pressure (rising) or vacuum line. Entities will include but not limited to air valves, reflux valves, scour valves and reducers.

Compulsory Attributes:

- i. Size – Nominal Diameter of pipeline valve is connected too. Reducer size shown as 250/150.
- ii. Open/Closed – Whether valve is open or closed under normal operating conditions.

Desirable Attributes:

- iii. Control – Manual or Automatic
- iv. Material – Valve Material
- v. Seat – Valve seat material
- vi. Pressure Setting - Pressure Setting in Pascal's
- vii. Valve off Direction – Direction to turn off valve i.e. Left/Right
- viii. Installation Date – Year in which the sewer asset was constructed or installed i.e. 2007

b. Manholes / Maintenance Shafts

Compulsory Attributes:

- i. Name - Identity label of manhole / Maintenance Shaft as per Appendix P
- ii. LidRL - Surface level of Lid, in metres A.H.D.
- iii. Depth - Depth to invert of Manhole / Inspection Shaft.
- iv. Type – Preformed or Cast In Situ
- v. Material – Manhole / maintenance shaft material PB or PE
- vi. Class – Classification rating.
- vii. Lid_Material - Material of lid.
- viii. Diameter – Inside diameter of manhole

Desirable Attributes:

- ix. Seal – Seal Type.
- x. Lining – Manhole lining.
- xi. Installation Date - Year in which the sewer asset was constructed or installed i.e. 2007.

c. Endcaps

Compulsory Attributes:

- i. Name - Identity label of endcap as per Appendix P.
- ii. IL - Invert level.

Desirable Attributes:

- iii. FSL – Finished Surface level above location of Endcap.
- iv. Installation Date - Year in which the sewer asset was constructed or installed i.e. 2007.

d. Sewer Pipes

- Gravity Sewer

Compulsory Attributes:

- i. Size – Pipe inside diameter.
- ii. Material – Pipe material.
- iii. Pipe Length – Plan length from end to end. Chamber dimensions are not to be included.. The length is recorded in metres.
- iv. USIL – Upstream invert level of pipe.
- v. DSIL – Downstream invert level of pipe.
- vi. Class – Pipe Class

Compulsory attributes if different to FNQROC standards:

- vii. Lining – Pipe lining material
- viii. Lining - Lining installation date

Desirable Attributes:

- ix. JTYPE – Joint type
- x. Location – Above or below ground, default below
- xi. Grade – Grade of pipe between manholes/ inspection shaft or endcap.

- Pressure (Rising) Sewer Mains

Compulsory attributes:

- xii. Pipe diameter
- xiii. Pipe material.
- xiv. Length – Plan length of pipe measured between pump stations, pits, bends, connection or discharge point.
- xv. Pipe material class

Compulsory attributes if different from FNQROC Standards:

- xvi. Year of Installation.

Desirable attributes:

- xvii. Pipe Protection
- xviii. Pipe lining material
- xix. JTYPE – Joint type
- xx. Location – Above or below ground, default below
- xxi. Depth of pipe – if not standard.

e. Storage / Overflow

Compulsory Attributes:

- i. Top of Well A.H.D
- ii. Depth - Depth to invert of Manhole / Inspection Shaft.

Required attributes if different from FNQROC Standards:

- iii. Invert level in metres of outlet pipe, A.H.D
- iv. Invert level in metres of inlet pipe, A.H.D
- v. Floor Slab Level and top of tank in metres, A.H.D
- vi. Well diameter
- vii. Overflow invert level

Desirable Attributes:

- viii. Identify asset i.e. Storage or overflow
- ix. Lid_Material - Material of lid.

- x. Capacity of well
- xi. Installation Date - Year in which the sewer asset was constructed or installed i.e. 2007.
- xii. Protection coating system used
- xiii. Real Property description

f. Pump and Lift Stations

Compulsory Attributes:

- i. Council Identification number.
- ii. Pump or Lift station.
- iii. Lid RL – Lid level in metres, A.H.D.
- iv. Depth – Difference between lid RL and Well Invert
- v. Well Invert – Invert level of well in metres, A.H.D.
- vi. Inlet invert – Invert level of inlet pipe in metres, A.H.D.
- vii. Overflow invert – Invert level of outlet pipe in metres, A.H.D.
- viii. Alarm RL – High level alarm in metres, A.H.D.
- ix. Invert Level of Rising Main in metres, A.H.D.
- x. Standby Start Level in metres, A.H.D.
- xi. Duty Start Level in metres, A.H.D.
- xii. Pump Stop Level in metres, A.H.D.
- xiii. Riser Pipe Diameter
- xiv. Inlet Level in metres, A.H.D.

Desirable Attributes:

- xv. The protection coating system used
- xvi. Real Property description
- xvii. Wet Well Diameter
- xviii. Inlet Diameter – Pipe inlet diameter.
- xix. Outlet Diameter – Pipe outlet diameter.
- xx. Diameter – Well Diameter or dimensions.
- xxi. Wet/Dry – Whether well is wet or dry.
- xxii. Capacity – Emergency Storage Capacity of well in hours.
- xxiii. Volume – Operational volume.
- xxiv. Pumps – Number of pumps.
- xxv. Flow – Design flow rate.
- xxvi. Start Frequency – Design starts per hour.
- xxvii. Installation Date - Year in which the sewer asset was constructed or installed i.e. 2007.
- xxviii. Reference Dwg – Drawing number for design of constructed sewer asset.

- xxix. Slab Level in metres, A.H.D.
- xxx. Underside of plug in metres, A.H.D
- xxxi. Pump Suction Level in metres, A.H.D.
- xxxii. Valve Pit Floor Level, Lid level and outside FSL in metres, A.H.D.
- xxxiii. The Level at Which the contributing sewerage system or sewer pumping station will overflow in metres. A.H.D.
- xxxiv. Rising Main Diameter
- xxxv. Incoming Sewer Diameter
- xxxvi. Capacity of wet well
- xxxvii. Maximum Operational Volume
- xxxviii. Cross-reference detail drawings and standard drawings
- xxxix. The protection coating system used
- xl. Real Property description

g. Valve Pit

Compulsory Attributes:

- i. Installation Date - Year in which the sewer asset was constructed or installed i.e. 2007.
- ii. Reference Dwg – Drawing number for design of constructed sewer asset

h. Property Connection Branch

Compulsory Attributes:

- i. Chainage – Distance from downstream manhole/inspection shaft
- ii. Type – Type as per standard drawing S3005
- iii. IL – Invert level of connection to sewer pipe, in metres A.H.D.

c) STORMWATER

Attribute Information is to be supplied for all new waste water assets which ultimately become the property and responsibility of Council. Attribute information is recorded in the drainage layer format and the various blocks as listed in the “As Constructed” Drawing Requirements stormwater drainage section.

a. Catch Drains

Compulsory Attributes:

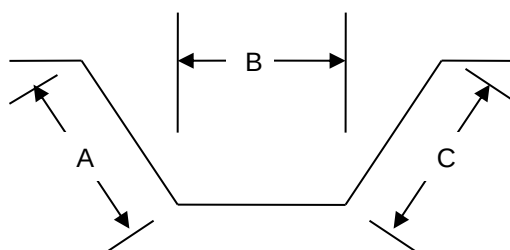
- i. Material – Construction Material.
- ii. Length – Slope length of drain
- iii. USIL – Upstream invert level.

- iv. DSIL – Downstream invert level.

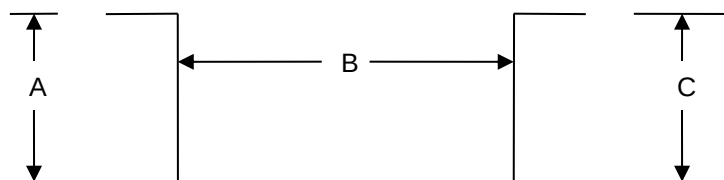
Desirable Attributes:

- v. Type – Type of Catch Drain i.e. lined, unlined and lined invert.
- vi. Shape – Shape of channel, Trapezoidal, Rectangular, V-Channel or Spoon
- vii. DIMA – Dimension A, average length of lined material, refer below.
- viii. DIMB – Dimension B, length of lined material, refer below.
- ix. DIMC – Dimension C, average length of lined material, refer below.
- x. Depth – Depth from channel invert to top of lowest embankment.
- xi. Thickness – Average thickness of channel lining.

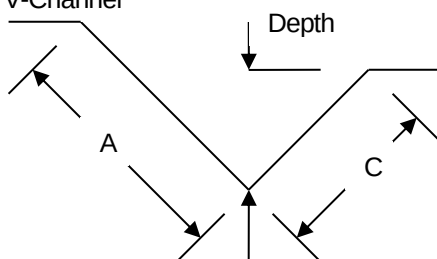
Trapezoidal



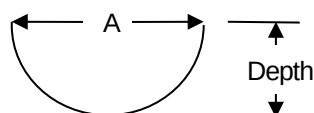
Rectangular



V-Channel



Spoon



Note: Where changes in cross sections occur, the relevant attributes should be noted on 'As Constructed' drawings

b. Stormwater Quality Improvement Device

Compulsory Attributes:

- i. Type – Type of SQID.
- ii. Manufacturer – Name of Manufacturer of the device.
- iii. LidRL – Surface level on the centre of the cover lid.

- iv. SumpIL – Invert level at the lowest point in the device.

Desirable Attributes:

- v. LidMaterial – Material from which the cover is constructed.
- vi. StorageRL- Level at which maximum storage capacity is reached.
- vii. Volume – Volume of material in cubic metres that the device is capable of holding.
- viii. DimensionA – Overall internal maximum width in millimetres.
- ix. DimensionB – Overall internal maximum breadth in millimetres, zero value if circular.
- x. InstallationDate - Year in which the drainage asset was constructed or installed i.e. 2007.

c. Detention / Retention and Bio-Retention Basins

Compulsory Attributes:

- i. Type – Type of Basin, Dry, Wet, Bio.
- ii. Capacity - Storage capacity in storage basin.
- iii. Weir – Surface level of weir in metres, A.H.D.
- iv. Length – Length of weir at recorded weir height.
- v. Material – weir Material, soil, rock etc.
- vi. Area – Area of filter material.
- vii. Filter – Type of filter material.
- viii. Filter_depth – Depth of media material.
- ix. Wall – Wall material type.
- x. Wall_volume – Volume in cubic metres of wall material.
- xi. UGStorage – Underground Storage or Drainage, Yes/No
- xii. InstallationDate - Year in which the drainage asset was constructed or installed i.e. 2007.
- xiii. Reference Dwg – Drawing number for design of constructed drainage asset.

d. Head and Wing walls

Compulsory Attributes:

- i. Apron – Yes/No, default no.
- ii. Wingwalls – Yes /No, default no.
- iii. Debris Trap – Yes/No, default no.
- iv. Floodgate – Yes/No, default no.

Desirable Attributes:

- v. Type – Precast, Cast In Situ, Stone Pitched.
- vi. Width – Length of headwall.
- vii. Height – Height of headwall.

- viii. InstallationDate - Year in which the drainage asset was constructed or installed i.e. 2007.

e. Kerb / Field Inlet Pits and Manhole / Access Chambers

Compulsory Attributes:

- i. LidRL - Surface level on the centre of the cover lid, grate or centre of kerb inlet.
- ii. IL – Invert level at the lowest point in the pit/ manhole or chamber.

Desirable Attributes:

- iii. Type – Type of Inlet
- iv. Dimension 1 – The overall internal width or diameter of pit.
- v. Dimension 2 – The overall height of a rectangular pit.
- vi. Cover Material – Cover Type.
- vii. Chamber Material – Construction material of the chamber.
- viii. Lintel No's – Number of lintels.
- ix. Grate No's – Number of grates.
- x. InstallationDate - Year in which the drainage asset was constructed or installed i.e. 2007.

f. Pipe / Culvert

Compulsory Attributes:

- i. Type – Pipe, Culvert or Slab Link Culvert.
- ii. Number – Number of pipes/culverts
- iii. Dimension 1 – The overall internal width for culverts or diameter of pipes.
- iv. Dimension 2 – The overall height of a culvert.
- v. Length – Slope length from end to end. Chamber dimensions are not to be included and pipe length shall be actual length, not plan length. The length is recorded in metres.
- vi. Material – Construction Material.
- vii. Class – Classification of pipe/culvert type.
- viii. USIL – Upstream invert level of pipe/culvert.
- ix. DSIL – Downstream invert level of pipe/culvert.

g. Pump Station

Compulsory Attributes:

- i. LidRL - Surface level on the centre of the cover lid.
- ii. SumpIL – lowest point in the well.
- iii. Reference Dwg – Drawing number for design of constructed drainage asset.

Desirable Attributes:

- iv.
- v. Material – Pump chamber material, concrete, masonry block etc.
- vi. Dimension 1 – The overall internal width for well.
- vii. Dimension 2 – The overall height of well.
- viii. Capacity – Capacity of well.
- ix. Volume – Operational volume.
- x. Installation Date - Year in which the drainage asset was constructed or installed i.e. 2007.

h. Tidal Flap**Compulsory Attributes:**

- i. Type – Type of Tidal Flap
- ii. Manufacturer – Manufacturer of the tidal flap.
- iii. Material – Construction Material.
- iv. Size – Dimension of tidal flap.

Desirable Attributes:

- v. Installation Date - Year in which the drainage asset was constructed or installed i.e. 2007.

d) ROAD

Attribute Information is to be supplied for all new road assets which ultimately become the property and responsibility of Council. Attribute information is recorded in the road layer as text or in the various blocks as listed in the “As Constructed” Drawing Requirements road section.

An “As Constructed” drawing is not acceptable if it does not include at minimum the compulsory attributes listed below.

a. Pavement / Surfacing**Compulsory attributes:**

- i. Each pavement Layers – Type of pavement material used, as per Main Roads Standard Specification.
- ii. Each pavement Layers_Depth – Depth of pavement material.
- iii. CBR – Sub-grade CBR test results, based on a 4-day soaked CBR test of the in situ sub-grade material upon which the pavement design was based.

Compulsory pavement attributes required if they vary from FNQROC Standard Construction requirements:

- iv. Surface – Type of wearing course material, Asphaltic Concrete, 2 Coat Bitumen, 80mm Pavers or Concrete.

- v. Surface_Depth – Depth of sealed road surface where applicable.
- vi. Reinforcement – Type of reinforcement used in concrete surfacing.

b. Pathways

Compulsory pathway attributes required only if they vary from FNQROC Standard Construction Requirements:

- i. Type – Material type, concrete, paved, Asphaltic Concrete
- ii. Depth – Depth of material.

c. Signs

Compulsory Attributes:

- i. MUTCD Code – Numbering system for sign specified by the Queensland Department of Main Roads in the Manual of Uniform Traffic Control Devices (MUTCD).
- ii. Name – Common name for the sign, i.e. street, give way stop etc.
- iii. Number – number of signs attached to the same supports.
- iv. Supports – Number of support posts the sign is attached to.

d. Roundabouts

Compulsory roundabout attributes required only if they vary from FNQROC Standard Construction Requirements:

- i. Type – Type of roundabout, concrete or vegetated.

e) PARKS / LANDSCAPING / STRUCTURES**a. Playground Equipment**

Compulsory Attributes:

- i. Type – Type of playground equipment, i.e. Swing, climbing frame, adventure playground etc.
- ii. Make – Manufacturer of the play equipment.
- iii. Model – Manufacturers model number for the play equipment
- iv. Installation Date - Year in which the play equipment was constructed or installed i.e. 2007.

b. Landscape Embellishments

Compulsory Attributes:

- i. Type – Descriptive comment on feature type.
- ii. Material – Construction material, if applicable.

- iii. Dimensions – Dimension of feature, if applicable.
- iv. Installation Date - Year in which the asset was constructed or installed i.e. 2007.

c. Buildings / Structures

Compulsory Attributes:

- i. Type – Type of building / structure
- ii. Roof – Roof material
- iii. Wall – Wall material
- iv. Floor – Floor material.
- v. Dim_A – Width of building in metres.
- vi. Dim_B – Depth of building in metres.
- vii. Height – Height to roof line in metres.

d. Tree

Compulsory Attributes:

- i. Type – Species name.
- ii. Name – Common Name.
- iii. Size – Trunk size.